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FACULTY OF ECONOMICS AND ADMINISTRATION

**Portfolio Modelling using the
Goals based Investment
Approach**

Master's thesis

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Abstract

This thesis constructed a portfolio using the goals based investment approach. The study used 10 historical assets prices spanning from April 12th, 2010, to March 12th, 2022. The study employed the goal-based investment approach to allocate resource across four mental accounts (subportfolio). In applying the goal-based framework, a case study was presented which was indicative of the numerous challenges (investment goals) faced by the investor. The portfolio optimization was carried out in the goal-based setting space and the Mean-variance space using the Monte Carlo engine.

Declaration

“I certify that I have written the master’s thesis “Portfolio modelling using the goal-based investment approach,” by myself under the supervision of Ing. Ludek Benada, Ph.D., and I have listed all the literary and other specialist sources in accordance with legal regulations, Masaryk University internal regulations, and the internal procedural deeds of Masaryk University and the Faculty of Economics and Administration”

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ATUAH, THEOPHILUS SAKYIAMAH

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List of Terms and Abbreviations

BPT	Behavioral Portfolio Theory
MV	Mean Variance
GBI	Goal Based Investing
MVO	Mean Variance Optimization
MPT	Modern Portfolio Theory
CAPM	Capital Asset Pricing Model
APT	Arbitrage Pricing Theory
VaR	Value-at-Risk
CVaR	Conditional Value at Risk
MDD	Maximum Drawdown
AGG	US Aggregate Bond Index
IFGL	International Developed Real Estate ETF
HYG	High Yield Corporate Bond

Introduction

In times past, investment professionals have modelled their portfolios premised on the ground-breaking work of Harry Markowitz. Markowitz's risk-based asset allocation became the building block on how portfolios would be built by investment professionals. These professionals had a duty to model the investors' goal by first decoding them into the language of finance, then appropriately determine an investment mix and finally manage these constructed portfolios (Boe and Ruud, 2018, p.1). The essence of the investment process is to come up with efficient portfolios that maximizes return for a stated amount of risk. The investor then selects a point on the frontier that connotes his tolerance for risk.

The tenet of Markowitz work hinges on diversification. The intuition of diversification is that by having a portfolio consisting of a variety of assets, investments that are underperforming are offset by those that are performing well leading to a more coherent pattern of return. In bear markets, investors benefit from diversification. This was mostly evident during the US bear markets from 2000 to 2002. Those investors who had not well diversified their portfolios and were concentrated in one sector were severely affected.

Despite some investors having well diversified portfolios those investors were not spurred the repercussion of the bear market as their well-diversified portfolios could not afford to meet their goals. The global financial crisis in 2008 further cast a slur on the benefit of diversification. It became apparent that diversification only produce desired results under a normal market condition as these results could not be replicated in a bear market. The investors had formulated investment strategies without a coherent goal and a coherent penchant for risk.

With the passage of time, Markowitz's theory has been exposed of some practical shortcomings that has to do with the theory's effect on the portfolio of investment agents. The first disengagement has to do with how academic theory and investors view investments. MPT is of the view that a stand-alone portfolio can hold multiple goals although these goals may have varying time horizons and risk profiles (Taylor, 2019). This practice in essence disregards the human tendency to segregate savings into separate accounts based upon the aim for those

savings. Largely, putting all investment in one bucket (portfolio) can lead to suboptimal investment decisions and investment returns. The other shortcoming has to do with how MPT explains risk. MPT explains risk as the standard deviation of returns. In essence, investors view risk as the failure to achieve a goal.

As a result of these failures, investment professionals have been forced to rethink their investment strategies to come up with a strategy that deviates from the traditional investment strategies of Markowitz. In their rethinking and to address these shortcomings, a new approach has surfaced that places the individual's goals front and center in constructing an investment strategy known as the goals-based investing (GBI) (Taylor, 2019).

1.1 Motivation

“Finance is not only about making money. It is about achieving our deep goals and protecting the fruits of our labor” as stated by the Nobel-laureate Robert Shiller (Parker, 2021, p.2). The Modern portfolio theory explains that investors are risk-averse and as such strive to maximize the risk-adjusted returns of their portfolios, however, in reality investors need capital to finance specific future investment goals. Investment therefore is to serve a specific purpose in life, with life goals being the purpose of the investment, and money being the conduit to achieve these goals. This presupposes that the act of investment is not done in a vacuum, it is underpinned by some goals in mind. Simply put, investing is at its core, goal based.

Once we recognize that investing is goal based, a suitable strategy is then developed premised on the cardinals of asset pricing and portfolio optimization that accurately reflects investors desired goals. These desired goals are translated into a mathematical optimization problem either to try to maximize the probability of achieving the desired goal by a given maturity date or minimize the anticipated shortfall (Krabichler & Wunsch, 2021, p.1).

Goal-based investing recognizes the importance of behavioral biases in investing. Thaler (1980) study threw more light on this. Thaler explained that investors are less concerned about the overall portfolio performance but are rather concerned with their mental accounts and how each mental account has its own unique objective and risk tolerance (Boe and Ruud, 2018, p.1).

Building on from Thaler's work, Shefrin and Statman (2000) proposed the Behavioral Portfolio Theory (BPT) where they redefined risk as the likelihood of failing to attain a mental account objective. Recognizing the insufficiency of the BPT framework Das et al., (2010) provided a superior framework with the aim of amalgamating the risk-is goal-failure structure of BPT with the Markowitz mean-variance framework (Parker, 2021, p.2). The findings from Das et al., (2010) showed that that mean-variance optimization and goal attainment probability maximization are statistically identical in most cases. The portfolio of the investor in this thesis will be constructed based on the Behavioral portfolio theory and then use Das et al., (2010) framework for the probability maximization strategy.

Boe and Ruud (2018) compared the mean-variance optimization approach with the goal-based optimization approach and found the goal-based approach providing superior performance to the mean-variance approach. This finding was supported by Aleksandra (2020) who also concluded that a coherent goal-based approach offers a higher probability of achievement than the Markowitz portfolio approach. However, none of these studies looked at how capital is to be allocated across multiple accounts to achieve an investor's goal. Parker (2020) filled this hole by providing a framework that sought to mathematically model how capital is to be allocated across different mental accounts to achieve an investment objective. This study differs from Parker (2020) work by using different investment mix with different capital market expectation and a different goal funding detail.

1.2 Purpose of study

The purpose of this study is to use an innovative approach to construct a portfolio premised on the goal-based investing approach and subsequently compare the approach with Markowitz mean-variance by comparing the probabilities of goal achievement in the goal-based space and the Markowitz Mean-variance space. The aim of this comparison will be to seek to determine which of the two approaches offers a superior solution of probability maximization. The study will also analyze how several optimized sub portfolios assigned to each goal in the goal-space

perform. The goal-based portfolio optimization will be modelled based on the framework of Parker (2021).

1.3 Significance of study

This thesis will seek to contribute to the Goal-based finance literature debate by seeking to affirm or disaffirm previous study on the subject matter. It will model the portfolio using different goal details. The study would be of great benefit to current and potential investors as well as wealth planners as it will provide a detailed perspective on how to allocate investors wealth across mental accounts. The findings in this thesis will offer an opportunity for further research using different capital market expectations as well as goal variables.

1.4 Limitation of study

This thesis assumed the use of liquid wealth only. It did not consider the impact of savings on the probabilities of goal achievement. Also, the study did not model the portfolio looking at the impact of tax rates. Moreover, the portfolio was not modelled based on the risk of goal set back or maximum drawdown rate as was the case with some of the previous studies. Lastly, this thesis assumes that there are no short-selling and leverage constraints.

1.5 Outline of thesis

The study is organized into eight chapters. Chapter one briefly describes the context in which the problem is situated. It further reveals existing gaps which the study sought to fill. The purpose, contribution and limitations of the study are also presented in this chapter.

Chapter two discusses in detail the theoretical framework underpinning the study. It further provides some empirical review of literature related to the study. Chapter three also provides details of the research approach, data and its sources, and the tools used in analyzing the data.

Chapter six presents the findings from the study and extensively discusses it relating it to the literature it reviewed in chapter two. Chapter seven provides a summary of the findings obtained in the study. It proceeds to make recommendations based on the findings and finally provides suggestions for future research. The R Studio code, which was used for modelling the portfolio can be obtained from the attachment in IS.

2 Literature Review

2.1 Theoretical Framework

This section will provide the necessary theoretical background upon which the strategies to implement the investment solutions will be premised.

2.1.1 Modern Portfolio Theory

The Modern portfolio theory is an inactive portfolio management method comprising 3 distinct portfolio theories which are largely premised on the portfolio risk and return profile to aid in the selection and construction of the portfolio (Vukovic and Bjerknes, 2017, p.14). The three theories are the Mean-Variance Analysis by Harry Markowitz, the CAPM independently developed by Treynor (1961), Sharpe (1964), Litner (1965), and Mossin (1966) as well as the Arbitrage pricing theory by Ross (1976). Among the contributors to the three theories, Harry Markowitz is broadly regarded as the father of modern portfolio theory. In 1952, Markowitz published a dissertation which has arguably been the bedrock of modern finance. In his dissertation, he developed a methodology of Mean-Variance Optimization (MVO) by which investment professionals can efficiently apportion assets based on two key factors: risk and return (Roche, 2016, p.3).

Markowitz's theory uses a complex mathematical process known as the Mean Variance Analysis to examine the covariance of asset groups within a particular portfolio and the effect a change in the asset apportionment will have on the total expected risk-return shape of the portfolio. The mean variance analysis relies on complex mathematical computation driven by intricate formulas to give credence to the concept's theoretical underpinnings (Mangram, 2013, p.59). To provide a simplified perspective of the key underpinnings of Markowitz's theory, several concepts have been developed drifting from the complex mathematical models which were advocated by Markowitz.

The CAPM makes a considerable input to the insight of the relationship between risk and return. The basic conjecture of this model is that the anticipated asset returns co-moves with the anticipated return on market portfolio with the difference in beta being able to substantially explain the cross-sectional difference in stock returns (Chiah, Chai, Zhong and Song Li, 2016). The beta measures the degree of co-movement between the security return and the return on the market portfolio as it also computes the systematic risk of an asset. CAPM was developed from different perspective by Sharpe, Litner and Mossin.

In deriving CAPM, Sharpe made great progress in expanding Tobin's work of efficient market frontier and Capital Market Line. A year on, Litner derived CAPM from the viewpoint of a corporation issuing shares. Finally, Mossin derived CAPM by explicitly specifying a quadratic utility function. Modern Portfolio theory has continued to evolve with several iterations to the works of Markowitz, Sharpe, Litner and Mossin. The main distinction between Markowitz's theory and CAPM is that Markowitz portfolio theory describe a set of actions that investors must follow in framing their portfolios while CAPM explain the actual behavior of investors in framing their portfolios. Nevertheless, these two concepts offer a theoretical framework to explain the nexus between projected return and risk.

The Arbitrage Pricing Theory (APT) is a build-up on the CAPM which presumes that assets are exposed to only market risk. The APT offers a linear relationship between numerous risk factors that may impact the asset. The instinct from the APT is that a security may be subjected to some firm specific risk factors such as the size of a firm and its book to market ratio and as such investors must think about before investing in that security. The theory is premised on the fact that returns on a single asset or a portfolio of assets can be predicted by means of the relationship between the asset in question and other several common risk factors. The returns an investor anticipates from a financial asset can however be exemplified in form of a linear function. The linear function reflects both market and economic independent factors as indicated earlier in modelling those functions.

2.1.1.1 Mean Variance Optimization (drawbacks)

Despite the tremendous contribution of the mean variance analysis of Markowitz, it has not been spared from criticism.

It has been argued that there is the likelihood for the mean variance analysis to document extreme weights because of a combination of extreme shorts with extreme longs which is generally triggered by the guesstimate errors in the mean return vector and covariance matrix (Haugh, Iyengar and Wang, 2016). These weights are extremely sensitive to small changes in the projected returns. This means that a minute increase or decrease in the projected returns of one asset can drastically modify the structure of the entire portfolio. The sensitivity of the weights tends to increase with the structure of the portfolio. For instance, as the asset classes increases in a portfolio, there is the likelihood for a larger estimation error for large portfolios. Best and Grauer (1991) expressed this drawback in the equation below

$$||w - \hat{w}|| \leq \epsilon ||\mu - \hat{\mu}|| \frac{1}{\gamma_{min}} \left(1 + \frac{\gamma_{max}}{\gamma_{min}}\right) \quad (2.1)$$

w and $\hat{w}$ connotes exact and projected optimal portfolio corresponding with the exact mean return vector μ and the projected mean return vector $\hat{\mu}$ while γ_{max} and γ_{min} relates to the largest and smallest eigen values of the covariance matrix. From the equation, it is evident that the sensitivity of the portfolio weights to errors in the mean vector return is directly related to the growth of $\frac{\gamma_{max}}{\gamma_{min}}$

Moreover, attention has mostly been devoted to the errors in the estimated means as it is easy to calculate neglecting the covariance errors which are normally present because of the heavy tails in the return distribution.

In overcoming these challenges, robust techniques have been developed such as the use of shrinkage estimators, robust estimators and Bayesian techniques such as the Black-Litterman framework introduced in the early 1990's.

Despite all the challenges, majority of investment strategy continue to be premised on the MPT.

2.1.2 Goal Based Investing

The investment difficulties of individual investors can largely be explained as a mixture of consumption or wealth goals linked to a set of dollar budget (current and future income) constraints as well as the risk budgeting like the maximum drawdown. It is imperative to note that the success or failure to achieve these goals does not entirely depend on the performance of a standalone fund nor a group of asset class. However, it hinges on how the performance of the portfolio dynamically cooperates with risk factors affecting the present value of the investment goals. The challenge, however, is how financial advisors can come up with an investment strategy that commensurate with the particular risk the individuals are faced with as against giving attention to the overall market risk (Deguest, Martellini & Milhau, 2021, p.1).

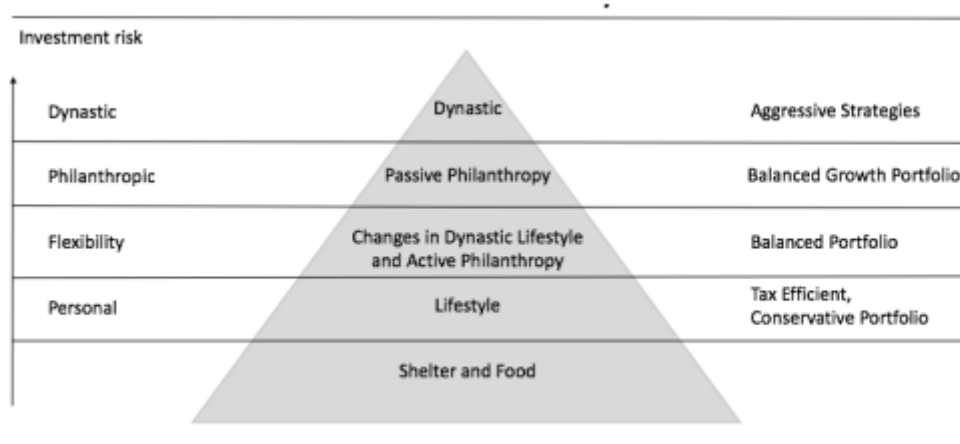
Goal Based Investing (GBI) is a predominantly new method in the wealth management ecosystem. Its main trust is investing to attain specific life goals. It involves the Investment firm's client using the financial markets to fund their specific goals within a stipulated period. GBI enhances an investment mix to minimize the prospect of failing to attain a minimum wealth level within a set period.

Goal based investors have several goals known as goal-space in which they allocate their capital among these goals by investing to achieve their stated goals. Taking inspiration from Maslow's hierarchy of needs, more important goals are prioritized over less important goals. Once capital is segregated based on an investor's goals, the portfolios are optimized to deliver the highest possibility of attaining each itemized lifetime goal. Segregating an investor's investment ensures precision when evaluating risk appetite for different sub portfolios. Das et al., (2011) were of the view that goal-based portfolio helps investors to frame more logically each goal, the period for each goal, and the penchant of risk for each goal.

Investors goals can be classified into three main groups; Personal (lifestyle requirements), philanthropic (personal values) and dynastic (children's future) (Brunel, 2011). Deguest, Martellini, Milhau, Suri and Wang (2015) also classified investors goals into three categories: essential goals (affordable and secured goals), important goals (affordable but non-secured goals), and aspirational goals (non-affordable and non-secured goals). According to the

Behavioral finance portfolio pyramid in figure 1 (Statman, 2004), shelter and food are the most basic and important needs to be satisfied first before considerations are given to other needs. The goals are arranged in layers where personal goals are ranked over philanthropic goals and dynasty. As a result of the diverse levels of importance, risk acceptance levels will also tend to vary.

Figure 1- The Behavioral Finance Portfolio Pyramid



Source: (Boe, Sverre and Ruud, 2018, Page 5)

Mladina (2016) posited that any sound goals-based asset distribution strategy should embrace these characteristics namely,

- Built on a portfolio theory
- Amalgamation of goals (liabilities) and time
- An innate definition of risk penchant
- Incorporation of recourse decision making and adaptive trade-of

2.1.2.1 Portfolio Theory

Portfolio theory is the cardinal point in comprehending asset distribution strategies, underpinning assumptions, and vital issues. There are likely to be consequences such as suboptimal outcomes in events an asset allocation strategy is not premised on a portfolio theory. Several theories have been offered in the academic literature to serve as a basis for asset

allocation. This includes Markowitz's Mean-variance analysis, and CAPM. However, none of these theories incorporates goals and time. Going forward, Intertemporal CAPM was put forward to breach the goals and time gap which were left unattended by previous theories (Cochrane, 2014).

2.1.2.2 Behavioral portfolio theory

The Behavioral Portfolio Theory was put forward by Shefrin and Statman in 2000. This theory examines how investors invest in reality. This theory is a deviation from the traditional finance theories which holds that investors select a diversified portfolio on the mean-variance efficient frontier. BPT rather assumes that investors construct their portfolios in stages or blocs. Each layer mirrors a varying risk and return profile. The goals of the investor are then used to assign between the varying layers. This presupposes that the BPT connects two issues, the creation of the portfolio and the shape of the securities (Shefrin & Statman, 2000). In building the portfolio based on the BPT, first, the goals of the investors are determined and ranked in an order of priority to determine how much resource is allocated to each layer. The next step is the asset allocation. This is done for each layer taking cognizant of the goal of the layer in question. If a higher return is required, then more speculative securities will be put into that bloc. What determines the number of securities in each bloc is the investors' appetite for risk. A risk-averse investor will put more into the bloc to mitigate the likelihood of not achieving his goals. This risk averse investor is likely to hold more cash positions to safeguard his investment so that he is not pushed to sell his assets when there is a need for liquidity. This theory provides a solid theoretical foundation on which goal-based investment strategies can be formulated

Goals and time

Every investor's goal is to ensure that the asset at his disposal can fund his lifetime financial goals. These goals are captured as liabilities on the lifetime balance sheet of the investors. If these assets' main purpose is serving the financial lifetime goals of the investor, then it is only prudent that investment strategies are goal based for multiple periods to accommodate the lifetime goals of the investors.

From a behavioral finance point of view, investors see their portfolio's as comprising diverse fundamental mental accounts sub portfolios, with each subportfolio representing a diverse goal. Diverse goals usually have divergent priorities with some being short term while others are long term. It therefore becomes necessary for the alignment of the risk profile of each subportfolio dedicated to fund the respective goals. Sharpe made a great contribution to this mental accounting debate by indicating that the overall portfolio is mean-variance efficient when mental account sub portfolios lie on the efficient frontier. The drawback of Sharpe's assertion is that standard mean-variance theory analysis is usually single-period optimal whereas investors goals are multi-period and funded over an investor's lifetime. It is therefore important to consider time and investors goal is coming up with any asset allocation strategy.

An innate definition of risk penchant

One of the daunting tasks of an investment officer is his/her ability to identify and integrate the risk tolerance level of an investor. The advantage of correctly identifying the investors risk level is the investment officer ability of designing a portfolio that resonates with the investors risk profile (Jansen, Kramer, & Boender, 2013). Risk can broadly be described as the chances that an expected outcome will not be achieved. For an investor, risk is the quantum of uncertainty the investor is prepared to admit regarding the future returns they assume from their investment. In Markowitz Mean-Variance Analysis, risk is explained as the standard deviation of expected return, whilst in the goal -based portfolio theory, risk is captured differently. In the goal-based theory, risk is explained as the likelihood of not achieving a set of stated goals which is how investors intuitively perceive risk (Brunel, 2011). Contributing to the behavioral finance and mean variance definition of risk, Das et al., (2010) stated that, in the mean-variance analysis, investors are mostly risk averse, whilst in the behavioral finance theory (goal-based portfolio), these investors may be risk seeking. Nevertheless, most of these concepts are closely related (Madlina, 2016). Efforts have thus been made in recent times to provide more-intuitive means to express risk penchant that can be deciphered into the language of portfolio theory.

Loss aversion is a cognitive bias where individuals favor avoiding losses over making gains. This behavior is possibly connected to marginal utility of wealth. Under normal condition, a

rational investor would contemplate risk to be the variance around the projected mean return of a well- diversified multi-asset class portfolio. However, loss aversion advocates otherwise. Loss aversion advocates that the investor will assign more weight to the negative returns than the implied variance around the projected mean return. Loss aversion has been captured in goal-based strategies to offer a more intuitive meaning of risk preference in selecting a portfolio. For instance, top ranked goals can be associated with the intertemporal risk-free asset to provide assurance of funding those goals should they arrive in time. In other instances, goal-based investing strategies to finance discrete goals can be chosen using shortfall probability as the explanation of risk.

It is imperative to differentiate between how to quantify risk and the activities to evaluate risk tolerance. According to (Kemp et al., 2018) the method of evaluating risk tolerance should be a detailed conversation between the investment advisor and the investor. Instead of the investment advisor basing the conversation on the investment horizon and risk tolerance, the advisor should step up and offer likelihood outcomes for the diverse risk levels; maximum loss, minimum loss, and projected loss. This will help the advisor to formulate a better strategy based on the goals of the investor.

Incorporation of recourse decision making and adaptive trade-of

An investment's future cannot be predicted with a degree of certainty. With the passage of time, anticipated outcomes are domineered by realized outcomes. The realized outcome can be below or above the anticipated outcome. As Goals evolve with time, recourse decisions will be made in the future premised on information available in the future. It is therefore imperative that Goals-based investing process would deliberately integrate recourse decision-making and adaptive trade-offs into the asset apportionment method.

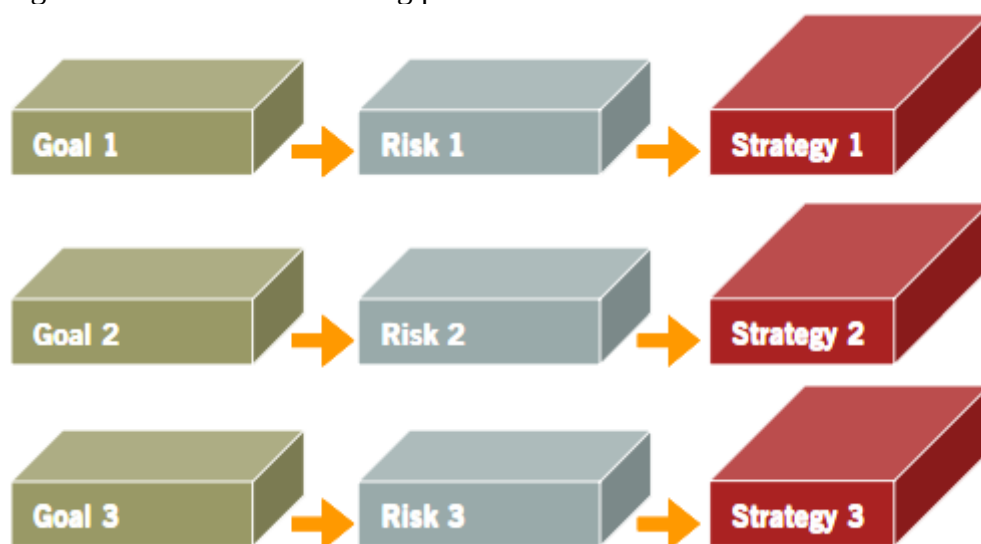
New information has the propensity to affect the shape of the asset allocation method and the tradeoffs that must be adapted to efficiently achieve the desired goals. The adaptive trade off can be reprioritizing the goal and altering the goals' threshold, time horizon, risk penchant and anticipated returns. Simply put, the process should be adaptive to changing conditions making it possible for investors to make knowledgeable personal and economic tradeoffs in the face of

fresh and relevant information. The trust of the goal-based investing approach is to be able to solve investors lifetime needs by providing optimum lifetime asset allocation and as such it is only prudent that any new information is thoroughly captured to yield a revised ideal lifetime asset allocation (Madlina, 2016).

2.1.2.3 Goal-based Investing process

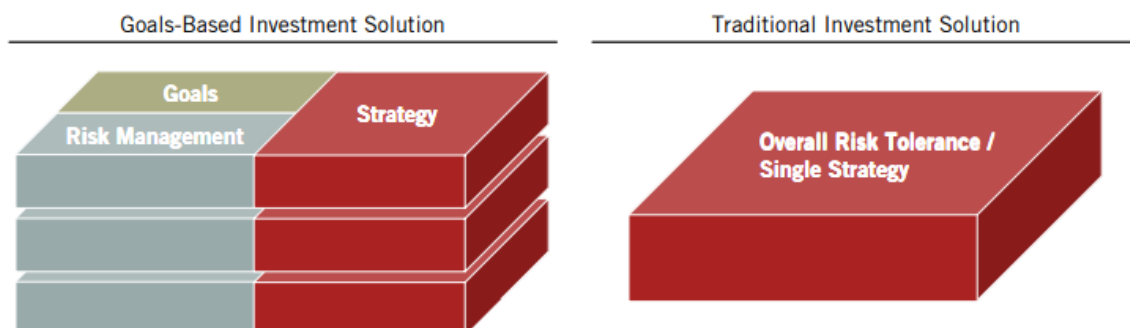
Figure 2 illustrates the goal-based investing approach. As can be seen from figure 2, the process connects investors goals to risk management objectives and investment strategies. Client goals are grounded on the investment portfolio's ultimate use, such as present day or future lifestyle demands. Risk management objectives are specified in terms of risk measures that best collaborate with the specific goal. Efficient Investment strategies are formulated corresponding to the client's goals and risk management objectives. This will lead to an asset pool with several sub-portfolio. Each sub-portfolio will be an amalgamation of a discrete goal, risk management objective and investment strategy. These Asset pools are then merged to generate a full investment solution.

Figure 2- Goals based Investing process



Source: Nevins (2003, P.13)

Figure 3- Traditional Investing process



Source: Nevins (2003, P.13)

In Figure 3, a comparison is made between a goals-based investment solution and a traditional investment solution. For the traditional investment procedure, efficient portfolios are fashioned employing standard investment theory. It starts by establishing the investor's overall risk tolerance through risk profiling. The established risk tolerance estimate is then linked to one of the efficient portfolios. Goals-based investing on the other hand has to do with establishing risk profile for each goal. This approach may, however, not be meaningful to investors without clear-cut goals, but when goals are explicitly laid out it has great benefits.

In sum, the goals-based portfolio approach comprises two separate elements. In one breath, it encompasses the stratification of investor likings into groups of goals that have comparable key characteristics, with priority standing and term structure of related liabilities, and on the other hand, it comprises the linking of these groups to enhanced performance or hedging portfolios having equivalent risk and return features.

The key challenge to a goal-based investing approach is its complexity which makes it difficult to execute. One of the key challenges to be addressed for successful execution has to do with the suitable level of customization and the procedure for developing the investment strategy. In its purest form, goal-based investing means customized strategy for each goal, but, when costs are well-thought-out, full customization becomes a mirage. A more utilitarian and well-fashioned strategy will be to come up with a group or class of investment products that can be paired with common investor goals. This then makes it easier to create customized solutions by selecting from the pack of products on goal-by-goal basis. For this method, investment products are

developed in line with risk management objective or performance. Another strategy to be considered can be allowing volatility in as much as it does not affect the investment outlook at the horizon date. This strategy is usually implemented for securities with well-established investment horizon, per chance concurring with a retirement date. In both strategies, risk management objective is paired with the investor goals for the suite of assets or securities.

Another challenge must also do with finding the optimal mix of investment. The ideal mix of investment is reliant on the quantum of resources assigned to the attainment of the goal. To address this challenge, the ideal mix of investment will first be obtained for discrete levels of wealth apportionment, then a Monte Carlo simulation will be used to obtain the maximal utility.

3 Empirical Framework

3.1 Mean-Variance Analysis

The mean-variance analysis presumes that once investors have complete information about an investment, they will make rational decisions. A key hypothesis of this model is that investors seek high returns while finding possible means to lower their risk. In Markowitz's (1952) view, the value of an asset is best examined by its mean, variance, and its correlation to other assets in a class of portfolio. Within the limitless options at the disposal of investors to construct a portfolio, balancing the risk and return features of the portfolio is what will lead to an optimal result. To reduce the investors exposure to risk, investors must take advantage of diversification by considering portfolios based on their entire risk other than putting so much energy on the individual assets without giving much attention to the overall performance of the portfolio (Brown, 2015). In sum, diversification helps to reduce portfolio risk.

All the notations are adaptation from Markowitz's (1952) publication. The mean variance analysis of Markowitz starts with a standard mean variance model.

In the mean variance framework, investor selects fractions of $Y_1, Y_2, Y_3, Y_4, \dots, Y_n$ invested in n securities, subject to the constraint:

$$\sum_{i=1}^n Y_i = 1 \quad (3.1)$$
$$Y_i \geq 0 \quad i = 1, \dots, n$$

The returns on the individual securities, given by $r_1, r_2, r_3, \dots, r_n$ are assumed to be jointly distributed random variables, henceforth, the portfolio return is expressed as:

$$R_p = \sum_{i=1}^n Y_i r_i \quad (3.2)$$

And the expected return on the entire portfolio is stated as:

$$E = \sum_{i=1}^n Y_i \theta_i \quad (3.3)$$

Where $\theta_i = E(r_i)$

The portfolio variance is expressed as

$$V = \sum_{i=1}^n \sum_{j=1}^n Y_i Y_j \vartheta_{ij} \quad (3.4)$$

Where

$$\vartheta_{ij} = E[(r_i - \theta_i)(r_j - \theta_j)] \quad (3.5)$$

Is the covariance between r_i and r_j and

$$\vartheta_{ij} = E[(r_i - \theta_i)]^2 \quad (3.6)$$

Is the variance of r_i .

3.2 The goal- based investing models

This study proposes that investors populate a goal space. The investors are assumed to assign values to each goal in the goal space creating a scale of preference. Axiomatically, it can be inferred that these investors favor higher likelihood of attainment to lower. Parker (2021) further asserted that the scale of preference follows Von-Neumann and Morgenstern's axiom of choice. In other words, preferences are complete, transitive, continuous and independent across the goal space. Given the above premise, it is conceivable to construct a coherent framework for the apportionment of limited resources.

The investors' goal space as adapted from Parker (2021) is stated as:

$$G: \{A, B, C, D \dots \dots \dots, N\} \in G$$

$A, B, C, D \dots \dots \dots, N$ are an N-number of distinct goals which strive for the investors limited initial resource pool, $\bar{\omega}$. Goal A is favored to goal B , and goal B is favored to goal C . Mathematically, a goal is expressed as a two-variable vector:

$$A = (W_A, t_A)$$

Where

W_A is the future wealth needed to finance the goal

t_A is the time within which the goal must be accomplished.

From the axiom of completeness, G can be ordered in ranks,

$$A \geq B \geq \dots \dots \dots \geq N$$

Mathematically, the function below is expressed:

$$v(.): A \geq B \Leftrightarrow v(A) \geq v(B). \quad (3.7)$$

This follows

$$\exists p: v(B) = v(A) \cdot p. \quad (3.8)$$

The value of p would be gotten using a certainty-equivalent approach. Let

$$v(A) = a: a \in \mathbb{R}^+ \quad (3.9)$$

For the purposes of this work, $a = 1$. The investor is then presented with a possibility: the attainment of goal B with certainty, or the attainment of goal A with probability p . The value of p will be methodically varied to find the indifference point such that,

$$v(B) = v(A) \cdot p \quad (3.10)$$

$$v(A) = a$$

Equivalently, we can express equation (3.10) as

$$v(B) = a \cdot p \quad (3.11)$$

The process will be repeated across the goal space and map out the value ratios one-to-one to G as

$$\{o, p, q, \dots \dots \dots, n\} \rightarrow \{A, B, C, D, \dots \dots \dots, N\} \quad (3.12)$$

The total value of the goal space, $v(G)$ will then be expressed as:

$$v(G) = a + ao + aop + \dots \dots \dots + aop \dots n. \quad (3.13)$$

From equation (3.13), the value ratios are the relative valuations of one goal for another. The equation permits the substitution of one goal for the other.

In instances of uncertainty, an investor devotes initial resources (wealth), ω , to the attainment of the goal. It can be deduced that $\omega < W$. In instances of certainty, $\omega \geq W$ and the goal would be accomplished with a degree of certainty.

To describe the likelihood of goal attainment given the preliminary resource, there must exist some function of the goal's variable factor and the preliminary input. This can be expressed as:

$$\exists \phi(\omega, W, t) = P_r[\omega_t \geq W] \quad (3.14)$$

From the initial equations, it was stated that, $\bar{\omega}$ is the preliminary resource pool from which the investor can fall on, in essence, equation (3.14) will be expressed as

$$\exists \vartheta: \omega = \bar{\omega}\vartheta \Rightarrow \phi(\bar{\omega}\vartheta, W, t). \quad (3.15)$$

From Markowitz's weighted sum formulation,

$$u(G) = \sum_i v(i)p(i) \quad (3.16)$$

Where, i is the index of goals and p is the likelihood of attainment for goal i . From the foregoing, equation (3.16) can be re-expressed as

$$p(i) = \phi(\bar{\omega}\vartheta, W, t) \quad (3.17)$$

It is therefore the objective of the investor to maximize utility by fluctuating the apportionment of the preliminary resources to each goal in the goals-space. This is expressed as:

$$\max_{\vartheta} \sum_i^N v(i)\phi(\bar{\omega}\vartheta_i, W_i, t_i) \quad (3.18)$$

Where ϕ is the probability of attaining a goal with the given input. A critical assessment of equation (3.18) reveals how narrow-minded the equation is. In the normal sense, an investor must make the most of inter-temporal utility in a continuous time. To overcome this narrow-mindedness, the equation will be re-optimized in discrete time such as quarterly or annually. A goal will be said to have been achieved when $\omega = W$. After accomplishing a goal, the goal is dropped from the goal space and the excess wealth is rededicated to the achievement of other goals in the space in terms of priority.

The annual return needed to attain a goal is expressed as

$$r(\vartheta) = (W/\bar{\omega}\vartheta)^{1/t} - 1 \quad (3.19)$$

Normality is then presumed which makes it possible to employ a cumulative distribution frequency to explain the likelihood of achieving the return vital for the accomplishment of the goal, as a function of resource distribution which is stated as:

$$\phi(\bar{\omega}\vartheta, W, t) = 1 - \Phi\left[\left(\frac{W}{\bar{\omega}\vartheta}\right)^{\frac{1}{t}} - 1; \mu, s\right] \quad (3.20)$$

$\Phi(\cdot)$ is the cumulative distribution function, μ is the portfolio expected return, and s is the anticipated standard deviation of the portfolio. The anticipated return and standard deviation of the portfolio are all functions of the portfolio's investment weight in the investment space, which is expressed as:

$$\mu = \sum_i \omega_i R_i \quad (3.21)$$

$$s^2 = \sum_i \omega_i^2 \sigma_i^2 + \sum_i \sum_{j \neq i} \omega_i \omega_j \rho_{i,j} \sigma_i \sigma_j \quad (3.22)$$

$\rho_{i,j}$ is the correlation of security i to security j , σ_i^2 is the variance of security i , and ω_i is the weight of security i . The investor will therefore maximize utility by fluctuating the apportionment of the preliminary resource pool to each goal in the goals-space and the weight of each security in the investment ecosystem. This is expressed as:

$$\max_{\vartheta, \omega} \sum_i^N v(i) \phi(\bar{\omega}\vartheta_i, W_i t_i) \quad (3.23)$$

This study will use both the logistic and the Gaussian distribution cumulative function.

3.2.1 Capital market expectation framework

As explained earlier, in a goal-based approach, risk is not standard deviation but the probability of failing to attain a stated goal while return is also explained as the return obtained over-and-above the minimum return needed to finance a goal. The goal here is to minimize the likelihood of goal failure and maximize returns above the minimum return. This is expressed in equation 3.24 and 3.25 as

$$\min \Phi(r_{req}/U, \sigma) \quad (3.24)$$

$$\max(U - r_{req}) \quad (3.25)$$

r_{req} is the return required annually to attain a goal, U is the portfolio expected return, σ is the portfolio standard deviation, $\Phi(\cdot)$ is the cumulative distribution function which assist in computing the percentage of likely returns which will be below the r_{req} . Moving on, we need to tweak the weights of the given assets to satisfy equation 3.24 and 3.25 which is our portfolio optimization goal. This will be done by incorporating forward-looking information as we seek to forecast future returns and volatilities.

To start with, the standard deviation of a portfolio according to the Markowitz portfolio theory is expressed in equation 3.26 as:

$$\sigma = \sqrt{\sum \sum \omega_i \omega_j \rho_{i,j} \sigma_i \sigma_j} \quad (3.26)$$

However, from a goal-based investing perspective, standard deviation is defined differently which has already been explained. To incorporate our forward-looking information, we substitute the historical standard deviation with implied volatility, in this way one can account for the markets future expected standard deviation. Equation 3.26 will thus be modified into equation 3.27 which is expressed as:

$$\sigma = \sqrt{\sum \sum \omega_i \omega_j \rho_{i,j} V_i V_j} \quad (3.27)$$

$V_i V_j$ is the implied volatility of the assets. The other variables have earlier been explained. The standard deviation formula in equation 3.27 is what will be used in equation 3.24.

After incorporating information on future implied volatility, we will proceed to incorporate information on future expected return. Resnick and Shoesmith (2002) study revealed that the US Treasury yield curve could predict economic environments. Parker (2018) thus used the 10-Year US Treasury and deducted the 3-Month US Treasury bill. This gives a fair picture of how the market behave in a recession. This will then be employed to examine how the various asset classes react in a recessionary environment.

Next, portfolio return will be relooked at to embrace the recession/non-recession likelihood suggested by the market. Portfolio return will be stated in equation 3.28 as:

$$U = (1 - P) \cdot \sum w_i q_i + P \cdot \sum w_i s_i \quad (3.28)$$

P is suggested probability yield-curve in a recessionary environment, w_i represent the suggested weight of the security, q_i connotes the return of the security in a non-recessionary setting, and s_i is also the security's return in a recessionary setting. Rightfully so, the expected return of the goal-based portfolio can be said to be the adjusted sum of the probability returns. Equation 3.27 and 3.28 will then be substituted into equation 3.24 and 3.25 as

$$\min \Phi(r_{req} | (1 - P) \cdot \sum w_i q_i + P \cdot \sum w_i s_i, \sqrt{\sum \sum \omega_i \omega_j \rho_{i,j} V_i V_j}) \quad (3.29)$$

$$\max \left(\left[(1 - P) \cdot \sum w_i q_i + P \cdot \sum w_i s_i \right] - r_{req} \right) \quad (3.30)$$

In goal-based setting, we usually exclude short-sales and leverage, this means that equation 3.29 and 3.30 will be subject to equation 3.31 which is expressed as:

$$\sum w_i = 1 \text{ and } 0 \leq w_i \leq 1 \quad (3.31)$$

The capital market framework explained is an adaptation and modification of Parker (2018, pp.11-15) framework.

Portfolio terminologies

Returns

According to Markowitz (1999), portfolio returns are the expected earnings produced from the invested asset or security. In this study, historical returns will be used as yield values cannot be studied for the future.

$$r_{it} = \frac{P_{it} - P_{it-1}}{P_{it-1}} \quad (3.32)$$

Where

r_{it} is the returns of security i at time t

P_{it} is the price of security i at time t

P_{it-1} is the price of security i at time $t - 1$

If dividend is added to the yield, we will have historical return as

$$r_{it} = \frac{P_{it} - P_{it-1} + D_{it}}{P_{it-1}} \quad (3.33)$$

Where D_{it} is the dividend for security i at time t

Equation (3.32) and (3.33) are mostly employed as a discrete method of calculating returns. In other instances, returns are calculated using logarithm when there is a high frequency data such as intra-day prices or continuous random variable. This thesis will compute the monthly returns based on equation 3.34 .The logarithm form is expressed as:

$$r_{it} = \ln \frac{P_t}{P_{t-1}} \quad (3.34)$$

In instances when an investor has several shares, historical returns is calculated as:

$$R_p = \sum_{i=1}^n Y_i r_i \quad (3.27)$$

Aside the above returns, geometric means and arithmetic means are sometimes used. The geometric mean is stated as:

$$R_I = [(1 + R_1) + (1 + R_2) + (1 + R_3) \dots (1 + R_n)]^{\frac{1}{n}} - 1 \quad (3.28)$$

$R_1 \dots \dots R_n$ are the returns of assets and n being the time of the investment horizon. The Geometric mean is normally used when the return amount is compounded and there is lot of volatility in the data set. It is usually used for longer time horizon. With this method, the actual amount invested must not necessarily be known.

The arithmetic mean is also expressed as:

$$R_i = \frac{[R_1 + R_2 + R_3 + R_n]}{n} \quad (3.29)$$

4 Risk Metrics

Several factors are considered before deciding on a particular risk metrics to use. It is imperative to consider the variations in mathematical properties, stability of statistical estimation, ease of optimization process, and most importantly, its acceptance by the industry ecosystem

4.1 Value-at-Risk

VaR has been one of the most widely used measures of risk across the financial industry. It gained popularity in the mid-1990s although its foundation can be traced to time past (Bustreo, 2013, p.2). The foundations of VaR can be traced to the mean-variance trade-off and optimization advocated by Markowitz (1952). VaR captures the risk of the total portfolio and has capacity to evaluate the possible shortfall in each portfolio (Bustreo, 2013, p.2).

VaR proclaims that for a specified confidence level $\alpha \in (0,1)$, the VaR is explained as the smallest number l , for which the probability of losing L is not greater than $(1 - \alpha)$ for losses higher than l . This value resonates with the percentile of the loss dispersal. The equation is expressed as:

$$VaR_\alpha = \inf \{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\} \quad (4.0)$$

$$\inf \{l \in \mathbb{R} : F_L(l) \geq \alpha\} \quad (4.1)$$

where F_L is the distribution function of losses.

Alternatively, VaR can be expressed as:

$$VaR_\alpha(Y) = F_Y^{-1}(\alpha) \quad (4.2)$$

When Y is normally distributed with the anticipated return and standard deviation, VaR will then be explained as:

$$VaR_\alpha(Y) = E(Y) + F_Y^{-1}(\alpha)\sigma(X) \quad (4.3)$$

For a stated portfolio, VaR can help an investor to know the anticipated loss over a period at a given confidence level. Statistically, it is the quantile of the allocation for variations in the value of the portfolio.

Let us assume a portfolio remain untraded for a specified period. Time is presumed to be measured in trading days. Let t_o be the current time, Mv_{t_o} be the current portfolio's market value. Let also assume that the future market value, Mv_{t_1} is unknown. As a random variable, a probability distribution can be linked to this conditional to time t_o . From this conditional distribution, the portfolio's market risk can be measured.

Let us contemplate a portfolio variation as:

$$\Delta MV = Mv_{t_1} - Mv_{t_o} \quad (4.4)$$

During the period of holding the portfolio. From equation 18, VaR can be stated as the solution to a non-linear equation expressed as:

$$P_r(\Delta MV \leq -VaR_\alpha) = 1 - \alpha \quad (4.5)$$

α is the level of confidence. VaR has been criticized over time. VaR is sub-addictive and difficult to control for non-normal distributions. The intuition of this shortfall is that a portfolio comprising sub portfolios may risk an amount which is equal to the total of the divided amounts risked by its sub portfolios. This sub-addictive challenge has been solved with the introduction of CVaR.

4.2 Conditional-Value-at-Risk

The concept of C-VaR was introduced by Rockafellar and Uryasev (2000). It is sometimes exchanged for expected shortfall. CVaR computes the amount of tail risk an investment portfolio has. It is obtained by taking the weighted average of the extreme losses in the tail of the distribution of likely returns. For VaR, it depicts the worst- case loss scenario linked with a probability and a time frame, while CVaR is the anticipated loss should that worst-case scenario be ever crossed.

To calculate the CVaR of a portfolio, VaR is first determined on the foundation of a stated time interval and a prearranged confidence level. In all instances the CVaR signify the mean loss level is the period loss is higher than VaR. To determine CVaR, the VaR and mean conditional exceedance of the Var are added. This means that CVaR will be always greater than VaR.

Mathematically, $CVaR_\alpha \geq VaR_\alpha$

CVaR is thus expressed as:

$$CVaR_\alpha(Y) = \frac{1}{1-\alpha} \int_\alpha^1 VaR_u(Y) du \quad (4.6)$$

For the incessant loss of Y with the associated F_Y distribution function and the confidence α , the

CVaR defined as

$$CVaR_\alpha(Y) = E(Y/Y \geq VaR_\alpha(Y)) \quad (4.7)$$

As stated earlier,

$$VaR_\alpha(Y) = F_Y^{-1}(\alpha)$$

Equation (4.7) can therefore be written as

$$\begin{aligned} CVaR_\alpha(Y) &= E(Y/Y \\ &\geq F_Y^{-1}(\alpha)) \end{aligned} \quad (4.8)$$

Granted that Y is a continuous variable, we can infer that:

$$CVaR_\alpha(Y) = VaR_\alpha(Y) + E[Y - \frac{VaR_\alpha}{Y} \geq VaR_\alpha] \quad (4.9)$$

But if, Y happens to be normally distributed, the equation will be stated as:

$$CVaR_\alpha(Y) = E(Y) + \frac{\varphi(F_Y^{-1}(\alpha))}{1-\alpha} \sigma(Y) \quad (4.10)$$

Where φ matches to the density function of the normal distribution

4.3 Risk optimal portfolio

4.3.1 Mean VaR portfolio

The mean-VaR portfolio model is premised on Markowitz Mean-Variance Analysis model. It substitutes the variance with the VaR as a measure of risk and it is framed as a multi-objective optimization problem where anticipated return is maximized and VaR is reduced to its barest minimum (Lwin, Qu, & MacCarthy, 2017, p.753).

The model starts with the VaR equation specified in equation (4.0) and (4.1) respectively as:

$$VaR_{\alpha} = \inf \{l \in \mathbb{R}: P(L > l) \leq 1 - \alpha\}$$

$$\inf \{l \in \mathbb{R}: F_L(l) \geq \alpha\}$$

mean -VaR is explained to be the difference between anticipated return and VaR. Mean-VaR can be looked at from two diverse viewpoint. One can either decide to determine the weights for a stated VaR to maximize portfolio return at a given level of confidence or choose to lessen the mean VaR for a stated portfolio at a given confidence level.

As a first step, it is presumed that there are N risky assets and wealth is apportioned among them. Let $Z_{\alpha} \Phi^{-1}(\alpha)$ connote the percentile of the loss circulation at the confidence level $\alpha \in (1/2, 1)$. Mean VaR will then be expressed as:

$$VaR_{\alpha}^m = Z_{\alpha} \sigma_W - \mu_p \quad (4.11)$$

$$\sum_{i=1}^n \omega_i = 1 \quad (4.12)$$

Equation (4.12) is a similar optimization problem to Markowitz's Mean Variance portfolio which was expressed earlier in equation (3.1) with a condition for the sum of the scales to be equal to one (Alexander & Baptista, 2002, pp. 1164-1166). Thus, the condition for the mean- variance efficient portfolio (Melton, 1972) is expressed as:

$$\frac{\sigma_{\omega}^2}{1/C} - \frac{(\mu_p - A/C)^2}{D/C^2} = 1 \quad (4.13)$$

Where the scalars A, B, C, and D are defined as

$$\begin{aligned}
A &= i' \Sigma^{-1} \mu \\
B &= \mu' \Sigma^{-1} \mu \\
C &= i' \Sigma^{-1} i \\
D &= BC - A^2
\end{aligned}$$

For a mean-VaR portfolio, the equation can be re-expressed to get the shape as:

$$\frac{[VaR^m + \mu_p/Z_\alpha]^2}{1/C} - \frac{(\mu_p - A/C)^2}{D/C^2} = 1 \quad (4.14)$$

Alexander and Baptista (2002) went further to examine changes in the shape of the efficient frontier for diverse confidence level (α). They resolved that; the optimization procedure has a solution only for a sufficiently high confidence level α . Mean-VaR portfolio can be minimized if the condition

$$\alpha > \Phi \left(\sqrt{\frac{D}{C}} \right)$$

Is met, otherwise there will be no solution. Once the criteria are met, the weight of the minimum Var portfolio at the given confidence level will be expressed as:

$$\omega VaR_\alpha^m = g + h \left[\frac{A}{C} + \sqrt{\frac{D}{C} \left(\frac{Z_\alpha^2}{C(Z_\alpha)^2 - D} - \frac{1}{C} \right)} \right] \quad (4.15)$$

Where the vectors g and h are defined as

$$g = \frac{1}{D} [B \Sigma^{-1} i - A \Sigma^{-1} \mu] \quad (4.16)$$

$$h = \frac{1}{D} [C \Sigma^{-1} \mu - A \Sigma^{-1} i] \quad (4.17)$$

in effect, the value of VaR_α^m agreeing to the weight vector (g and h) of the minimum-VaR portfolio α for the stated level of confidence will be stated as:

$$VaR_\alpha^m = Z_\alpha \sqrt{\frac{Z_\alpha^2}{C(Z_\alpha)^2 - D}} - \left[\frac{A}{C} + \sqrt{\frac{D}{C}} \left(\frac{Z_\alpha^2}{C(Z_\alpha)^2 - D} - \frac{1}{C} \right) \right] \quad (4.18)$$

As discussed earlier, minimum-VaR portfolios lies on the efficient frontier of the mean-variance portfolio, if any exist. By way of reasoning, one can also infer that, all portfolios lying on the efficient frontier can be said to be minimum -VaR portfolios with some form of confidence level. One can determine the values of the confidence levels α_{MV} from the relationship

$$\alpha_{MV} = \Phi \left(\sqrt{\frac{D}{C}} + \frac{D^2/C^3}{(\bar{r} - A/C)^2} \right) \quad (4.19)$$

4.3.2 Optimal drawdown portfolio

Taking a cue from the maximum drawdown, Chekhlov, Uryasev and Zabarankin (2005) proposed optimal draw-down portfolios. Their aim was to attain weight solutions regarding portfolio's drawdown. Investment managers usually put in place measures to avoid large portfolio drawdowns or drawdown over a longer period.

The drawdown of a portfolio is evident from equation 42 is the difference between the maximum uncompounded portfolio value before time t and the portfolio's value at time t . Three functional risk measures can be inferred from the definition: maximum drawdown (MaxDD), average drawdown (AvDD), and conditional drawdown at risk (CDaR). CDaR relies on the given confidence level α . CDaR is expressed as:

$$CDaR(\omega)_\alpha = \min_{\zeta} \left\{ \zeta + \frac{1}{(1-\alpha)T} \int_0^T [D(\omega, t) - \zeta]^+ dt \right\} \quad (4.20)$$

ζ is the threshold value for the drawdowns which is only surpassed by $(1-\alpha)T$. As a caveat, if the number of observations is not an integer, we will conclude on ζ being the weighted average of drawdowns. CDaR will serve as a risk functional and not a risk measure as was the case with CVaR which has explained as a risk measure. About optimizing a portfolio's drawdown, these risk functionals will serve as a disparity limitation for a static share of wealth at risk. For

instance, one could demand the MaxDD be pegged at 12% of the capital such that: $MaxDD(\omega) \leq v_1 C$, where $0 \leq v_1 \leq 1$ and C are the wealth/capital accessible. In similar function, AvDD and CDaR can also be expressed as $AvDD(\omega) \leq v_2 C$ and $CDaR(\omega) \leq v_3 C$ respectively. The portfolio optimization will then be presented in discrete terms with the aim of maximizing the annualized average portfolio return. It is expressed as:

$$R(\omega) = \frac{1}{dc} y'_T \omega \quad (4.21)$$

d represent the number of years in the time interval (0, T). Chekhlov et al. (2005) presented three linear program formulations to optimize a portfolio so that MaxDD, AvDD, and CDaR will be restricted by a portion of wealth. The attempt to maximize the portfolio's return regarding restricting the maximum draw down is expressed as:

$$P_{MaxDD} = \arg \max_{\omega \in \Omega, u \in \mathbb{R}} R(\omega) = \frac{1}{dc} y'_T \omega \quad (4.22)$$

$$u_k - y'_k \omega \leq v_1 C, \quad (4.23)$$

$$u_k \geq y'_k \omega$$

$$u_k \geq u_{k-1}$$

$$u_0 = 0$$

where u symbolizes the maximum portfolio values up to time k with $1 \leq k \leq T$.

if the portfolio is optimized with the aim of restricting the average drawdown while maximizing the average annualized return, the first set of disparity restrictions will be substituted with the discrete analogue stated in a continuous time resulting in:

$$P_{AvDD} = \arg \max_{\omega \in \Omega, u \in \mathbb{R}} R(\omega) = \frac{1}{dc} y'_T \omega \quad (4.24)$$

$$\frac{1}{T} \sum_{K=1}^T (u_k - y'_k \omega) \leq v_2 C, \quad (4.25)$$

$$u_k \geq y'_k \omega$$

$$u_k \geq u_{k-1}$$

$$u_0 = 0$$

For the CDaR linear program formulation to optimize a portfolio, two additional supplementary variables will be added to the loose variables (u). The first of the two is the starting point draw down value ζ which relies on the confidence level α , and the second being the $(T \times 1)$ vector Z constituting the weak starting point exceedances. It is represented as:

$$P_{AvDD} = \arg \max_{\omega \in \Omega, u \in \mathbb{R}, Z \in \mathbb{R}, \zeta \in \mathbb{R}} R(\omega) = \frac{1}{dc} y_T' \omega \quad (4.26)$$

$$\zeta + \frac{1}{(1 - \alpha)T} \sum_{K=1}^T Z_K \leq v_3 C, \quad (4.27)$$

$$Z_K \geq u_k - y_k' \omega - \zeta, \quad (4.28)$$

$$Z_K \geq 0$$

$$u_k \geq y_k' \omega$$

$$u_k \geq u_{k-1}$$

$$u_0 = 0$$

The MaxDD and AvDD has some shortfall. Pfaff (2016) explained that the MaxDD and AvDD might fail to capture portfolio losses accurately. This is because portfolio's which strictly follow the maximum drawdown principle may be adversely affected by a single worst draw down whilst the Average drawdown idea is indecisive about the actual size of the maximum portfolio's drawdown. The good thing, however, is that the CDaR can mitigate these shortfalls to a certain dimension.

4.4 Evaluating portfolio performance

4.4.1 Sharpe Ratio

Sharpe Ratio (SR) is the most widely used measure of risk adjusted performance. It was coined by William Sharpe in 1966. He defined it as the measure of excess return per unit of risk. Mathematically, it is expressed as:

$$SR_i = \frac{E(R_i) - R_f}{\sigma_i} \quad (4.29)$$

Where $E(R_i)$ is the anticipated return of security i , R_f is the risk-free rate and σ_i is the standard deviation of security i . SR is usually construed as a reward-to-risk-ratio, where both risk and return are expressed in one measure. From an investors point of view, a rising anticipated returns and a falling standard deviation are good omen for investment. The higher the SR, the better the risk-adjusted performance. SR offers a more accurate risk-adjusted performance when the assets in the portfolio are uncorrelated

SR is usually scale variant, that is does not rely on the amount invested, nor the degree of the investor's appetite for risk, but rather rely on the investment horizon. SR may not be an accurate measure of risk-adjusted performance for a goal- based investor as it has no direct link with the utility (goal) of the investor. SR is concerned with the total wealth at the end of the investment horizon, whereas the goal-based investor is much concerned with the specific wealth to fund his or her specific goals within the goal space. For SR to work effectively for a goal-based investor, the goals must be expressly linked to the expected return and its variance. That is an attained goal should e directly linked to the expected return.

SR is normally used when securities are normally distributed, as the allocation of returns can then be completely explained by its mean and standard deviation. However, in instances when the returns are not normally distributed then it is imperative to rely on non-normally distributed performance measures.

4.4.2 Modified Sharpe Ratio

The Modified Sharpe Ratio (MSR) was introduced to deal with the limitations of SR. Gregoriou and Gueyie (2003) proposed a modified Sharpe Ratio which can measure risk-adjusted performance of non-normally distributed returns. They replaced the standard deviation in the denominator with the Modified Value-at-Risk (MVaR). MVaR can account for return asymmetries (Skewness) and fat tails (kurtosis) at a prescribed level of confidence α . MVaR is thus expressed using the Cornish Fisher expansion as:

$$MVaR = \mu - \left(Z_c + \frac{1}{6} (Z_c^2 - 1)\tau + \frac{1}{24} (Z_c^3 - 3Z_c)k - \frac{1}{36} (2Z_c^3 - 5Z_c)\tau^2 \right) \sigma \quad (4.30)$$

where μ is the mean of the investment, σ is the standard deviation of the investment, τ is the security's skewness, κ is the investments kurtosis and Z_c is the critical value of the normal standard distribution at a $(1 - \alpha)$ threshold. By replacing standard deviation with MVaR, MSR will be expressed as:

$$MSR = \frac{E(R_i) - R_f}{MVaR} \quad (4.31)$$

MSR has gained popularity among practitioners due to its precision and ease of computation. Nevertheless, MSR has its own drawbacks. MSR is restrained by the domain of definitions (see Chernozhukov, Fernandez -Val and Galichon, 2010). The finance literature has not been able to vividly document the domain of defining the Cornish Fisher expansion which leaves the use of the Cornish Fisher expansion to situations where the returns are almost normally distributed.

4.4.3 Maximum drawdown

It is explained as the maximum cumulative loss from peak to trough before a new peak is established. Investors are often caught up in liquidity trap making it difficult to secure funds to finance their goals normally after a downside market decline. This sometimes forces the investors to sell of other investment positions to finance more prioritized goals under hostile market conditions. In scenarios of heavy drawdown, normal risk metrics such as volatility, Expected Shortfall and Value-at-Risk at the end of the stipulated investment period becomes less meaningful, which makes practitioners (investment advisors) turn their attention to the maximum drawdown (Goldberg & Mahmoud, 2016). Mathematically, it is expressed as:

$$MDD = \max_{t \in [0, T]} \{W(\omega, \tau)\} - W(\omega, t) \quad (4.32)$$

Where $\{W(\omega, \tau)\}$ is the portfolio value before time t and $W(\omega, t)$ is the portfolio's value at time t . MDD only considers large losses but not take into consideration the frequency of the losses. This makes it difficult to measure the time it took for the investment to be recovered. As a rule, a low drawdown is preferred to a higher drawdown as a low drawdown is an indication that losses realized from the investment were minimal.

4.4.4 Sterling Ratio

Sterling Ratio is a performance metric used in assessing the performance of a risk-adjusted investment return. It explains how risk capital is efficiently distributed. Sterling Ratio helps to quantify an investment performance by making a comparison between periodic returns and periodic drawdowns. Unlike Sharpe and Sortino Ratio which defines risk as standard deviation and deviation of negative returns respectively, Sterling Ratio defines risk as the investment average maximum drawdown. A higher Sterling ratio is an indication that a fund may have a bigger return relative to the anticipated level of risk. It is expressed as:

$$SR = \frac{\text{Annual portfolio return} - \text{Annual risk} - \text{free rate}}{\text{Average Maximum Drawdown}} \quad (4.33)$$

4.4.5 Sortino Ratio

The Sortino Ratio is used to assess the extra return for each unit of downside risk. It is similar to the Sharpe Ratio; the difference is that Sortino Ratio differentiate between harmful volatility from the total volatility by using the standard deviation of negative returns (downside deviation) as the denominator. It is expressed as:

$$SR = \frac{\text{Expected portfolio return} - \text{Annual risk} - \text{free rate}}{\text{standard deviation of negative portfolio returns}} \quad (4.34)$$

4.4.6 Omega Ratio

It is a weighted risk-adjusted performance measure which provides a comparison between the chances of winning to the chances of losing. Omega ratio takes into the consideration the overall return distribution when evaluating performance. It provides valuable insight into the performance of the portfolio in situations when investment returns are not normally distributed. Unlike the other risk-adjusted performance measures like the Sharpe Ratio which only considers only volatility, Omega Ratio can account for all moments of the distribution, that is the third and

fourth momentum effect, skewness and Kurtosis which is a useful tool in comparing performance of investments. As a rule of thumb, a higher ratio is an indication that asset provides a higher degree of gains over losses for the minimum acceptable returns. It is expressed as:

$$\Omega = \frac{\int_d^k (1 - F(y)) dy}{\int_c^d F(Y) dy} \quad (4.35)$$

Where F represent the cumulative distribution function of the returns, d represent the tolerable return which can be categorized as a gain or loss.

5 Methodology

This chapter provides a detailed step by step approach of the procedures and methodology that were employed in modelling a portfolio based on the Goal based investment strategy. It stipulates a detailed explanation on the research approach and the research design employed and further presents the data used, its sources and the estimation technique used in executing the analysis. The estimation technique is carried out in R package. The R script used is an adoption of Parker (2020) script with some modifications to reflect the changes in the variables used in this study.

5.1 Data

The study used a data set representing four Asset class, namely Stock, Commodity, Fixed Income, and Alternative Investment. The data for the study was sourced from Yahoo Finance. The stock Assets comprises Apple, Euro Stoxx 50, and Google. On the other hand, two assets namely Gold, and Oil were used to represent the commodities class. In proxying for Fixed Income, the US Aggregate Bond, and the High Yield Corporate Bond were chosen. Finally, three Assets namely International developed Markets, Emerging Markets and CASH were chosen as a proxy for the alternative Investment. The study used the four different asset class to ensure the portfolio is well diversified to reduce the total risk of the investment portfolio. In essence the diversified portfolio will help to reduce the impact of volatility on the overall portfolio. A detailed explanation of the assets is explained next.

APPLE

Apple specializes in software, consumer electronics and online services. It is ranked among the five biggest American information technology companies. As a result of the company's expansion into other areas, its prices continue to increase offering reasonable returns to investors. The company's competitive edge is its brand name. Investors are drawn to Apple due to the resurgence of the iPhone, dividends, fewer shares, fashion trend, and the Warren Buffett seal of

approval (Vena, 2021). The goal-based investor hopes to enjoy some good dividends over the period it will invest in Apple.

Euro Stoxx 50 (FEZ)

The Euro Stoxx 50 index is the foremost blue-chip index developed by STOXX, an index provider for Europe's Eurozone. The index is a free-float market capitalization founded in 1998 to provide a blue-chip representation of the major sectoral leaders in the zone. The index consists of 50 stocks obtained from 8 Eurozone countries namely, France, Belgium, Spain, Germany, Italy, Ireland, Spain, and Netherlands. The index serves as a license to many financial institutions to serve as an underlying to derive an array of investment products such as options, futures, and structured products (Qontigo, 2022)

Google

Google is an Alphabet incorporated class A stock. Its parent company is Alphabet which happens to be the world's third biggest generating revenue company. Google is one of the most impressive stocks in the 21st century. The stock was launched in 2004 at a price of \$50 but has since skyrocketed to \$2703.38. Google has a moat in the internet industry as its subscribers continues to increase over the years because of delivering better results at a faster speed as compared to its competitors.

Gold

It is a precious metal which serves as a safe haven for most investors in times of economic turbulence. In times of economic turbulence, its prices rise or holds its value as most governments and investors turn to it as a safe haven against uncertainty. On the other hand, in stable environments, its price drops. It is used as a hedge against inflation. Gold was included in the asset class to provide a hedge against uncertainties to help the goal-based investor achieve his investment goals with some degree of certainty.

Oil

Oil interacts with almost every facet of an economy as a consumer good or raw material for industries. Oil was included in the portfolio to serve as a speculative asset, a diversifier, or a hedge against other positions. The proxy used in representing oil was the West Texas Intermediate.

US Aggregate Bond Index (AGG)

The Bloomberg US aggregate Bond Index (AGG) is a broad base market capitalization-weighted bond index denoting the US investment grade bonds traded on an intermediate term basis. The index is used as a benchmark by investors for measuring the overall performance of the US Investment grade Bond market. Moreover, the index helps in tracking Mortgage-Backed Securities, government debt, and Asset Backed Securities to mimic the universe of investments that meet some laid down criteria

High yield corporate bond (HYG)

This type of bonds offers a higher rate of return owing to a higher probability of default. The proxy for representing the High Yield Corporate Bond was the iShares iBoxx \$ High Yield Corporate Bond ETF. This index tracks the performance of the Markit iBoxx USD liquid high yield index comprising U.S dollar denominated, high yield corporate bonds.

MSCI Emerging Markets Index (EM)

This is an alternative investment (ETF) with an objective of tracking the performance of an index comprising large and mid-capitalized emerging market equities. This asset has been included in the portfolio to help the goal-based investor diversify internationally while at the time benefitting from long-term growth. The goal-based investor will have access to over 800 + emerging market equities. The index was formally launched on 4th July 2003.

International Developed Real Estate ETF (IFGL)

This index seeks out to trail the investment performance of an index consisting of real estate stocks in developed markets outside U.S jurisdiction. This index will give the investor access to an array of international real estate equities and Real Investment Trusts (REITs). By including this in the portfolio, the goal-based investor will be exposed to the real estate industry in developed markets like Asia, Europe, and Canada.

Meta Financial Group, Inc. (CASH)

Meta Financial Group, Inc works through its banking subsidiary, Meta Bank to offer solutions for banking and non-banking businesses. The bank's primary business is attracting deposits and investing the deposits in the bank's lease and loan portfolio, while also providing prepaid cards and other financial solutions to businesses. The bank is intermittently engaged to sell loans, such as government guaranteed loans, and consumer credit product loans to third party buyers. By adding this to the portfolio, the investor will benefit from the interest on the loans advanced by the bank though sometimes not much.

The study used Daily data which was later converted to monthly data spanning from April 12th, 2010, to March 12th, 2022. The study period was chosen to reflect the various turmoil the financial industry has been exposed to over the years.

5.2 Steps in the Goal -based portfolio modelling approach.

The goal of the thesis is to construct an optimal portfolio premised on the goal-based investment approach and then compare the probabilities of achieving the goal between the goal-based strategy and the Mean Variance constraint.

The goal-based portfolio modelling approach will be carried out in the steps explained below

1. The first step is to determine the goal variables namely the initial wealth (resource pool), time horizon, and the future expected wealth.
2. The second step is to develop a capital market expectation for the investment universe namely correlations, volatility forecast and expected return forecast.
3. The final step is running a standard optimizer with a goal-based utility function.

5.2 Determining the Goal variables

Case study

The investor is a young investor aged 30 years with a potential to climb the wealth ladder. The investor has four separate goals with different profiles. I assumed Gaussian normal distributions throughout the case study. The investor is presented with the following characteristics:

Four goals, $G = \{A, B, C, D\}$, with funding requirements and time horizons of:

- I. \$4,500,000 needed in 30 years, $A =$
 $(\$4,500,000; 30);$
- II. \$2,000,000 needed in 20 years, $B =$
 $(\$2,000,000; 20);$
- III. \$800,000 needed in 10 years, $C =$
 $(\$800,000; 10);$
- IV. \$70,000 needed in 5 years, $D =$
 $(\$70,000; 5);$

The goal is expressed as a two-variable vector. The next step is to determine the goal values. It is determined that the value ratios are $v(A) = a = 1$, $b = 0.45$ and $c = 0.5$, $d = 0.58$. The investor has a total resource pool of $\pi = \$3,500,000$, split between his personal bucket and market bucket (Deguest et al., 2021).

Table 1- Initial Resource pool available to the Investor

Personal bucket		Market Bucket	
Residence	\$2,000,000	Equities	\$2,800,000
cash	\$500,000	Bonds	\$700,000

Source: Author's construct.

Table 1 shows that the investor has \$6,000,000 at his disposal to carry out his investment needs. However, investor would want to use his market bucket to finance his goals. This means that the

initial wealth dedicated to the achievement of the goal is \$3,500,000. The investor will however be willing to draw from his personal bucket (Residence) in events the probability of achieving his goal A is below 90% which is the most important goal. The acceptable threshold of success is presented in Table 2

Table 2- Acceptable threshold of success

Goals	Threshold of success
Goal A	90%
Goal B	83%
Goal C	80%
Goal D	82%

Source: Authors construct

From Table 2, the investor is willing to accept an achievement probability of 90% for Goal A, 83% for Goal B, 80% for Goal C, and 82% for Goal D.

Financial Goals

Goal A: Retirement income of \$4,500,000 in 30 years. The investor expects to retire from active work in 30 years and would require \$4,500,000 to finance his retirement needs, such as consumption and other important needs

Goal B: college funding of children's education sufficient for a four-year college education and 2-years post graduate education in 20 years' time

Goal C: Purchase a house worth \$800,000 in 10 years' time

Goal D: Go on a family vacation in 5 years' time worth \$70,000.

The goal details are presented in Table 3.

Table 3- Goal details

	GOAL A	GOAL B	GOAL C	GOAL D
Value Ratio	1	0.45	0.5	0.58
Funding Requirement	\$4,500,000	\$2,000,000	\$800,000	\$70,000
Time Horizon	30	20	10	5

Source: *Author's construct*

5.3 Capital Market expectation

Before developing the capital market expectation, the study will calculate the assets returns, standard deviation, and correlation matrix of returns to produce a covariance matrix. The asset returns will be computed using the difference of the log of the previous closing price with the current day's closing price. The returns (mean) will then be used in forecasting the expected returns.

Expected returns and volatility will be generated based on Parker (2018) market-based forecasting model which has been explained in the literature review. The inputs are the average returns, forecasting period.

The next step is to develop the capital market expectation. The capital market expectation is shown in Table 4. Euro Stoxx 50 is expected to have the highest return with oil expected to have the lowest return. In terms of volatility, Oil is expected to be the most volatile asset among the 10 asset class followed by Google and Emerging markets. As expected, CASH will be the least volatile asset. The essence of the capital market expectation is to find the combination of investments which provides maximal probability of goal achievement for each possible level of resource pool (ϑ), for each goal in the goal space. As indicated earlier in the literature review, this will be done for several level of granularity; that is, it will be tested across whole percentage points (1%, 2%, 3%, 4%....100%). Simply put many potential portfolio allocations will be simulated for each value of resource pool and each goal. This will be proceeded by documenting

the best investment mix that provides the maximum level of goal attainment likelihood (for each value of resource pool).

Table 4- Capital market expectation

Asset	Return Forecast	Volatility Forecast
Apple	0.090	0.150
Euro stoxx 50	0.110	0.160
Google	0.120	0.170
US Agg Bond	0.040	0.050
Gold	0.060	0.190
Oil	0.040	0.320
Emerging Markets	0.090	0.170
International Developed	0.070	0.150
US High Yield	0.060	0.090
Cash	0.010	0.001

Source: Authors construct

After finding the optimal investment mix, the study proceeded to find the mix of goal apportionments that maximizes utility. This was done by generating several resource pool allocations and matching the optimal investment mix with the resource pool allocation, then return the combination with the maximum utility. The subportfolio apportionments are then matched up with the appropriate theta allocation, and the subportfolio aggregate portfolio allocations are returned.

The mathematical representation can be found in the literature review from Equation 3.7 to Equation 3.23.

5.4 Running the Monte Carlo optimizer

The Monte Carlo simulation will be carried out in two phases. The aim is to be able to rationally distribute resources both within and across the four stated goals.

Phase 1: Optimal within goal allocation

The first step is to adjust the goal allocation and find the optimal investment portfolio and its characteristics for each level of across-goal allocation. This uses a non-linear optimization engine to find optimal portfolios. I started by cataloging the various possible across-goal allocations, (0% to 100%)

The next step is to iterate through each potential goal allocation to obtain the investment weights that offers the maximum probability of goal achievement. These weights will be specific to each goal, so they will be log into a matrix, where each row corresponds to a potential goal allocation.

Phase 2: Optimal Across-Goal Allocation

The characteristics of the within-goal allocations will be used to obtain the best across-goal allocation. To begin with, a matrix of simulated goal weights will be built, and then return the utility for each simulated portfolio.

5.5 Goal based investing With Mean-Variance Constraints

Following the norm of goal-based investors constraints, the study assumes that there is no short-sale and leverage constraints (Parker, 2021; Boe et al., 2018, and Das et al., 2021). The optimal mean-variance weights are calculated. The Mean-Variance weight will be calculated for both within goal and across goal allocation as was done in the case of the goal-based approach explained earlier.

After finding the optimal mean-variance weight, both within and across goal, based on Monte Carlo optimizer, the study will move on to obtain the last point on the mean-variance efficient frontier, for portfolios with less expected return than that being offered by the maximum on the frontier. These optimal weights will be identical with mean-variance optimization.

The final step is to compare probability of achievement for goals-based approach and mean-variance constraint approach.

The monthly log returns were computed in excel and then imported into R for the other analysis. A detailed explanation of the procedure is explained in the R script. The R script will be uploaded separately in IS.

6 Results

This section of the thesis explains how the methods, analytical procedures, techniques, models specification and estimation approach discussed in the literature review and methodology are applied to the data generated to achieve the research objectives. The results from the data analyses are displayed in tables and graphs where appropriate. The chapter proceeds with commentary on the key findings.

Descriptive statistics

The descriptive statistics of the monthly log returns are presented in Table 5. From the Table, all series exhibit positive average returns with different magnitude. A careful observation shows that the equity assets had higher average returns than the other class of assets, with Apple being the asset which recorded the highest average returns. In terms of volatility (standard deviation), Gold was documented as the most volatile asset among the Asset class used for this study. Gold had a standard deviation of 25.6% over the period. Several reasons have been put forward to explain why Gold is more volatile than the other assets. Gold is seen as a safe- haven (Salisu et al., 2021) where investors turn to in times of geopolitical tensions, equity market turmoil and a weakened dollar, and an increased in expected deflation (Lucey & Li, 2015). These events usually lead to a sharp change in prices resulting in higher volatility in gold prices.

Generally, most of the assets are heavy-tailed and leptokurtic as evidenced by their excess kurtosis values. The effect of this excess kurtosis is that the goal-based investor will occasionally experience extreme returns either positive or negative, which is more excessive than the normal + or – three standard deviations from the mean that is projected by the standard distribution of returns. It is therefore not surprising as Gold had the highest kurtosis value further validating the highest standard deviation it recorded and the being the asset with the maximum returns of 213.7% and the minimum returns of -211.9%. Nevertheless, Gold is an important asset to be included in the portfolio.

In terms of skewness, there is a mix of both left and right skewness, with majority of the assets being left skewed.

Table 5- Monthly returns Summary statistics

	Apple	Euro.50	Google	AGG	Gold	Oil	EM	IFGL	HYG	CASH
nobs	143	143	143	143	143	143	143	143	143	143
Min	-0.203	-0.198	-0.141	-0.026	-2.119	-0.568	-0.197	-0.251	-0.106	-0.912
Max	0.194	0.183	0.184	0.027	2.137	0.546	0.151	0.132	0.081	0.426
1 Quat.	-0.024	-0.040	-0.026	-0.004	-0.018	-0.046	-0.031	-0.021	-0.005	-0.027
3 Quat	0.073	0.046	0.059	0.009	0.029	0.055	0.033	0.033	0.014	0.073
Mean	0.021	0.004	0.016	0.002	0.004	0.001	0.002	0.004	0.004	0.014
Media n	0.026	0.008	0.015	0.002	0.003	0.013	0.005	0.008	0.005	0.017
Sum	3.002	0.507	2.330	0.341	0.519	0.121	0.306	0.593	0.597	1.948
SE M	0.006	0.005	0.006	0.001	0.021	0.010	0.004	0.004	0.002	0.011
LCL M.	0.009	-0.007	0.005	0.001	-0.039	-0.018	-0.007	-0.004	0.001	-0.008
UCL M.	0.033	0.014	0.027	0.004	0.046	0.020	0.011	0.013	0.008	0.035
Var.	0.006	0.004	0.004	0.000	0.066	0.013	0.003	0.003	0.000	0.017
Stdev	0.075	0.061	0.066	0.010	0.256	0.116	0.053	0.051	0.021	0.129
Skew.	-0.281	-0.242	0.186	0.018	0.063	-0.886	-0.322	-0.983	-0.626	-2.566
Kurt.	-0.140	0.538	-0.212	0.055	63.301	9.214	1.337	4.048	5.212	18.169

Source: Author's construct

Correlation matrix for Assets

Table 6 provides the correlation measure among all the assets. From the Table, one can observe both significant and insignificant correlations among the assets, albeit zero correlations between oil and Euro Stoxx 50. International Developed Real Estate ETF (IFGL) and Emerging markets exhibited the strongest positive correlation of 0.78 as well as High Corporate Yield bond (HYG) and International Developed Real Estate ETF (IFGL) also with a positive correlation of 0.78. All the stocks showed a strong positive correlation among themselves which was not surprising as they all had similar characteristics.

All the Alternative investments were positively linked to each other which was also expected as well as the Fixed income assets also being positively correlated. However, unlike the other asset class which were positively related, the commodities were negatively related. This negative

correlation between oil and Gold differs from previous studies which found a positive correlation between Gold and oil (Madani and Ftiti, 2021). Nevertheless, this positive correlation has not always been the case. There have been instances where crude price was down for more 35% while at the same time gold spot prices were up by over 19% (Meredith, 2020). Also, from the Table, US aggregate bond and CASH exhibited the highest negative correlation of 0.14

Table 6- Correlation matrix of Assets

	APPL	FEZ	GOOG	AGG	Gold	Oil	EM	IFGL	HYG	CASH
APPL	1	0.4	0.4	0.12	0.07	0.04	0.4	0.33	0.4	0.06
FEZ		1	0.54	-0.01	0.03	0	0.77	0.77	0.72	0.14
GOOG			1	0.03	-0.01	-0.06	0.42	0.5	0.46	0.08
AGG				1	0.12	-0.12	0.06	0.27	0.23	-0.14
Gold					1	-0.01	0.1	0.09	-0.03	0.02
Oil						1	0.03	0.05	-0.02	0.12
EM							1	0.78	0.75	0.13
IFGL								1	0.78	0.11
HYG									1	0.14
CASH										1

Source: Author's construct

Covariance matrix between Assets

After computing the correlation between the assets, the covariance matrix was computed from the correlation matrix. The results are displayed in Table 7. From Table 7, there is a mix of significant positive, negative and zero covariance among the assets. CASH exhibited zero covariance with all the other nine assets which means there is no predictable relationship between CASH and all the nine other assets. A careful look at the Table reveals that some other assets exhibited zero correlation with other asset. The highest covariances recorded were between the asset and itself. For instance, the covariance between Apple and Apple, or the covariance between Euro Stoxx 50 and Euro Stoxx 50.

Those Assets which are positively related means that once there is an increase in an asset there will be a corresponding increase in the other asset. On the other hand, the negatively related once

show that the assets move in different direction, that is an increase in the return of asset connotes a decrease in the returns of the other asset. It must be stressed however, that the covariance does not show the strength of the positive or negative relationship among the variables, but rather the correlation helps to explain the strength of such relationship.

Table 7- Asset covariance matrix

	Appl	Eur 50	GOOG	AGG	Gold	Oil	EM	IFGL	HYG	CASH
Appl	0.023	0.010	0.010	0.001	0.002	0.002	0.010	0.007	0.005	0.000
Eur 50	0.010	0.026	0.015	0.000	0.001	0.000	0.021	0.018	0.010	0.000
GOOG	0.010	0.015	0.029	0.000	0.000	-0.003	0.012	0.013	0.007	0.000
AGG	0.001	0.000	0.000	0.003	0.001	-0.002	0.001	0.002	0.001	0.000
Gold	0.002	0.001	0.000	0.001	0.036	-0.001	0.003	0.003	-0.001	0.000
Oil	0.002	0.000	-0.003	-0.002	-0.001	0.102	0.002	0.002	-0.001	0.000
EM	0.010	0.021	0.012	0.001	0.003	0.002	0.029	0.020	0.011	0.000
IFGL	0.007	0.018	0.013	0.002	0.003	0.002	0.020	0.023	0.011	0.000
HYG	0.005	0.010	0.007	0.001	-0.001	-0.001	0.011	0.011	0.008	0.000
CASH	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Source: Author's construct

Optimal within Goal Allocation

The first step is to adjust the goal allocation and find the optimal investment portfolio and its characteristics for each level of across-goal allocation. This uses a non-linear optimization engine to find optimal portfolios. The weights of the allocation are then calculated. These are the starting weights to help us know the percentage of holding each asset will have in the portfolio.

Asset starting weight

The assets starting weights are exhibited in Table 8. As indicated earlier these weights are based on the Behavioral portfolio theory. From the Table, High Yield corporate bond exhibited the highest percentage of holding in the portfolio. The portfolio will comprise of about 23.87% corporate bonds. Google followed suit as the asset with the second highest percentage holding. It returned a weight of 20.17%. This was followed by oil with a weight of 16.80%. Among the assets included in the portfolio to serve the goals of the investor, CASH had the lowest

percentage points of 0.48%. Table 8 does not reveal the nature of the investor with respect to each goal. The nature, that is the risk profile of the investor will be revealed in the optimal sub-portfolio weight. The other weights can be seen from the Table.

Table 8 – Assets starting weight

Asset	starting weight
Apple	0.0807
Euro stoxx 50	0.0337
GOOG	0.2017
AGG	0.0533
Gold	0.0326
Oil	0.1680
EM	0.0196
IFGL	0.1671
HYG	0.2387
CASH	0.0048

Source: *Author's construct*

Initial wealth allocation to each asset

After finding the optimal starting weight, the investor together with the adviser will then apportion the initial resource (\$3,500,000) among the 10 different assets. The breakdown is provided in Table 9.

From Table 9, \$835,275 will be invested in the High Yield Corporate Bond, \$705,970 will be invested in Google stock, \$587,844 will be invested in oil, \$584,780 will be invested in the International Developed Real Estate ETF. As seen from the Table, CASH will receive the least investible dollars of \$16,701. The other investible dollars can be seen from the Table.

Table 9- Initial Asset weight and Initial resource allocation

Asset	starting weight	Initial Allocation
Apple	0.0807	\$ 282,443
Eur 50	0.0337	\$ 117,860
GOOG	0.2017	\$ 705,970
AGG	0.0533	\$ 186,467
Gold	0.0326	\$ 114,143
Oil	0.1680	\$ 587,844
EM	0.0196	\$ 68,516
INT Dev	0.1671	\$ 584,780
HYC Bond	0.2387	\$ 835,275
CASH	0.0048	\$ 16,701

Source: *Author's construct*

6.1 Optimal subportfolio weight

As explained in the literature review, the goal-based investor has different risk-return profile for each goal. Risk management objectives are specified in terms of risk measures that best collaborate with the specific goal. Efficient Investment strategies are formulated corresponding to the client's goals and risk management objectives. The asset allocations based on each goal are presented in Table 10.

For Goal A, that is Retirement income in 30 years' time, the investor will invest 96% of the initial wealth dedicated to Goal A in equities. In terms of the specifics, 46% of Goal A's initial wealth will be invested in Google, 34% will be invested in Euro Stoxx 50, 14% will be invested in Apple. The remaining 6% is invested in Gold to diversify the portfolio. Goal A received heavy investment in equities as the investor expects to profit from the rising prices in the 30 years investment horizon. The investor has the luxury of time to be able to accommodate the volatilities in the equity market. This investor will be deemed as an aggressive investor who seeks for higher returns in the face of higher risk. As with the BPT, because the investor is

expecting a higher return to be able to achieve Goal A, he added more speculative assets in this case stocks to his portfolio.

For Goal B, the investment approach is like that of Goal A. The investor is seeking for higher returns while at the same time, is careful not to have a higher probability of goal failure. Goal B is the children's college education in 20 years' time. From Table 6.6, the investor allocates 29% of Goal B's initial wealth to investing in Google, 23% in Euro Stoxx 50, 13% in Apple, 9% in Gold and 2% in oil. As a risk-taking measure, the investor decides to invest 24% of the initial wealth in U.S Aggregate Bond Index which has low risk profile as compared to stock. Goal B investor can thus be termed as semi-aggressive investor.

For Goal C, the future required wealth is \$800,000 in 10 years. From the Table, the investor believes that the ideal investment to help realize the goal of the portfolio will be investing 100% in CASH. The investor allocates all the initial wealth apportioned to Goal C to CASH which is the asset with the least return and risk. The investor believes that the returns from investing in CASH although small will be enough to help realize the goal of that portfolio. From these characteristics, the portfolio will be termed as a conservative portfolio.

The last goal is Goal D. Goal D had a future required wealth of \$70,000. From Table 6.6, as it was with the Goal C, the goal-based investor will channel all his initial resource into CASH. An Investment in CASH is the ideal asset allocation to help achieve the Goal D. Portfolio D can also be said to be a conservative portfolio. In the eyes of BPT, Goal D investor can be said to be a risk-averse as he has all his holdings in cash positions to avoid the potential of having to sell some assets when there is the need for more liquidity.

Table 10– Optimal subportfolio weight

	Apple	Eur 50	GOOG	AGG	Gold	Oil	EM	IFGL	HYG	CASH
A	0.14	0.34	0.46	0.00	0.06	0.00	0.00	0.00	0.00	0.00
B	0.13	0.23	0.29	0.24	0.09	0.02	0.00	0.00	0.00	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
D	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00

Source: *Author's construct*

Optimal Goal weight

To be able to examine the probability of reaching each goal, I first calculated the optimal goal weight. The optimal goal weight is presented for the Goal-based investor and the Goal based investor with mean variance constraint. From Table 11, the optimal weight to deliver the maximum utility (goal achievement) to the investor are 29 for Goal A, 47 for Goal B, 22 for Goal C and 2 for goal D. a careful observation shows that the mean variance constraint investor also has similar optimal goal weight as the goal-based investor without minimum variance constraint.

Table 11- Optimal portfolio weight

Goal	Optimal Goal weight (GB)	Optimal MV constraint weight
Goal A	29	29
Goal B	47	47
Goal C	22	22
Goal D	2	2

Source: *Author's construct*

Optimal Aggregate portfolio performance

The results of the optimal aggregate investment portfolio are presented in Table 12. The optimal aggregate portfolio is based on the initial weight of each asset and the probability of achieving each goal. The highest optimal aggregate investment portfolio is 0.98. The rest are displayed in the Table.

Table 12- Optimal aggregate Investment portfolio.

Asset	Optimal aggregate portfolio
Apple	0.1334
Eur 50	0.2934
GOOG	0.3828
AGG	0.1236
Gold	0.0765
Oil	0.0102
EM	0.0000
IFGL	0.0000
HYG	0.0000
CASH	0.9800

Source: Author's construct

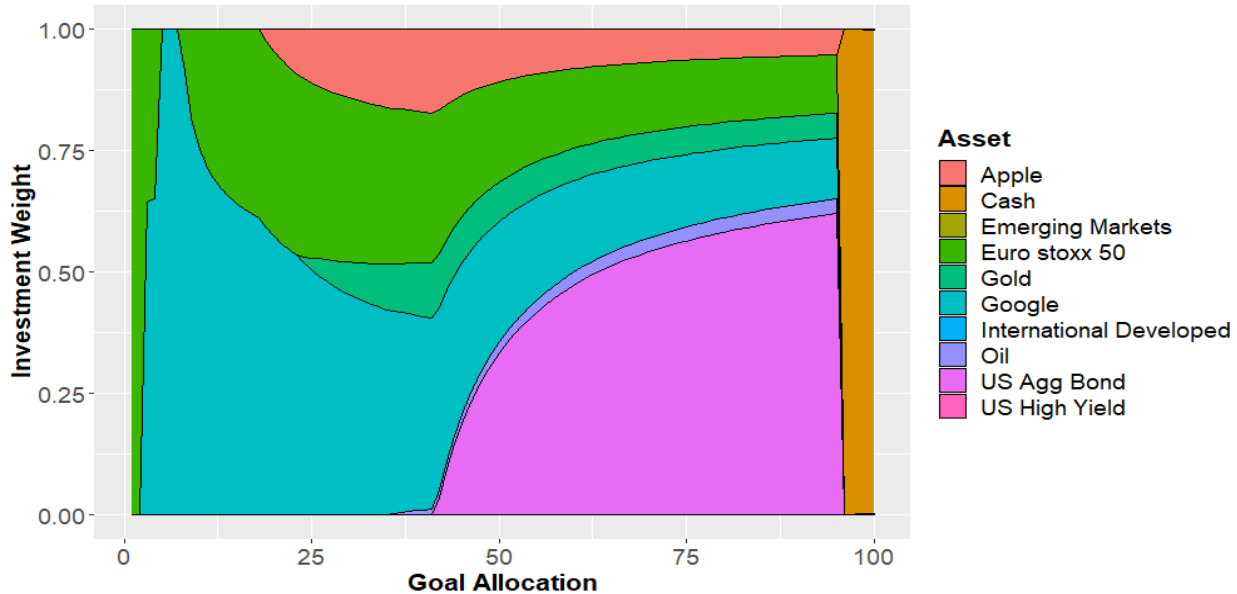
6.2 Investment Allocation as a function of Goal Allocation

The study further presents how the sub portfolios are allocated in relation to the across goal allocation in the goal-based framework.

Goal A's subportfolio allocation as a function of Goal A's across-goal allocation

Figure 4 indicates how Goal A's subportfolio are allocated in relation to Goal A's across-goal allocation. Basically, Figure 4 explains how Goal A's subportfolio is allocated in substituting Goal A for other goals. This framework is based on the goal-based approach as opposed to the mean-variance approach.

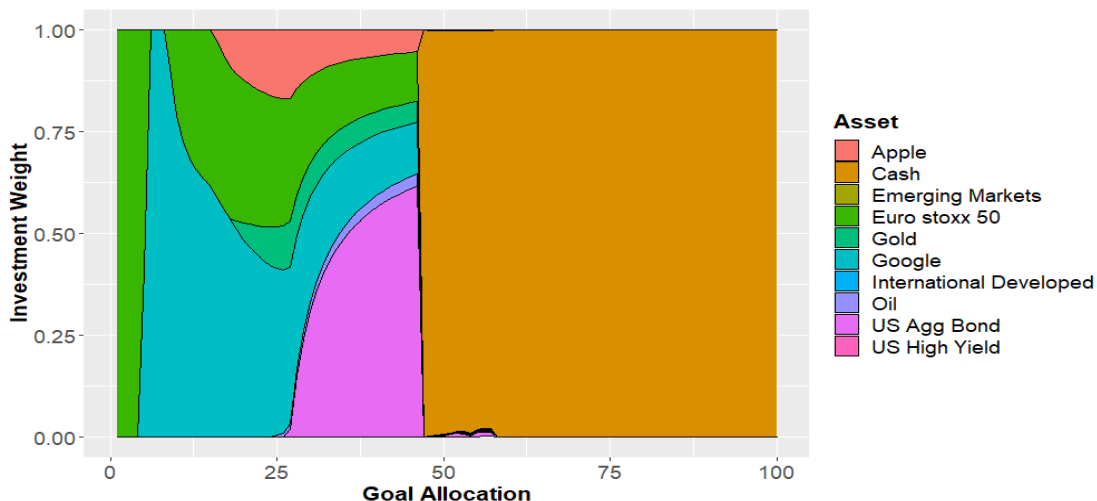
Figure 4- Goal A's subportfolio allocation as a function of Goal A's across-goal allocation



Source: R Studio

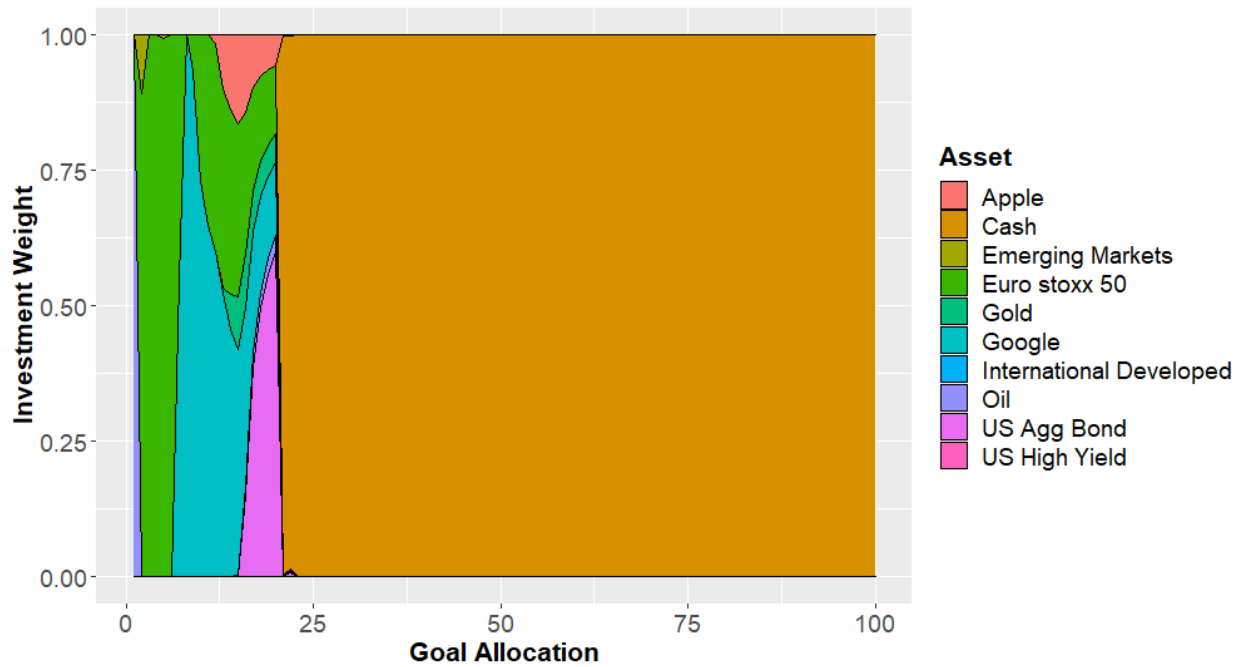
Figures 5,6, and 7 also represent a subportfolio allocation as a function of its across goal allocation. These figures depict how a Goal's subportfolio is allocated in connection with the Goals across goal allocation which has been explained earlier to mean substituting one goal for the other.

Figure 5- Goal B's subportfolio allocation as a function of Goal B's across-goal allocation



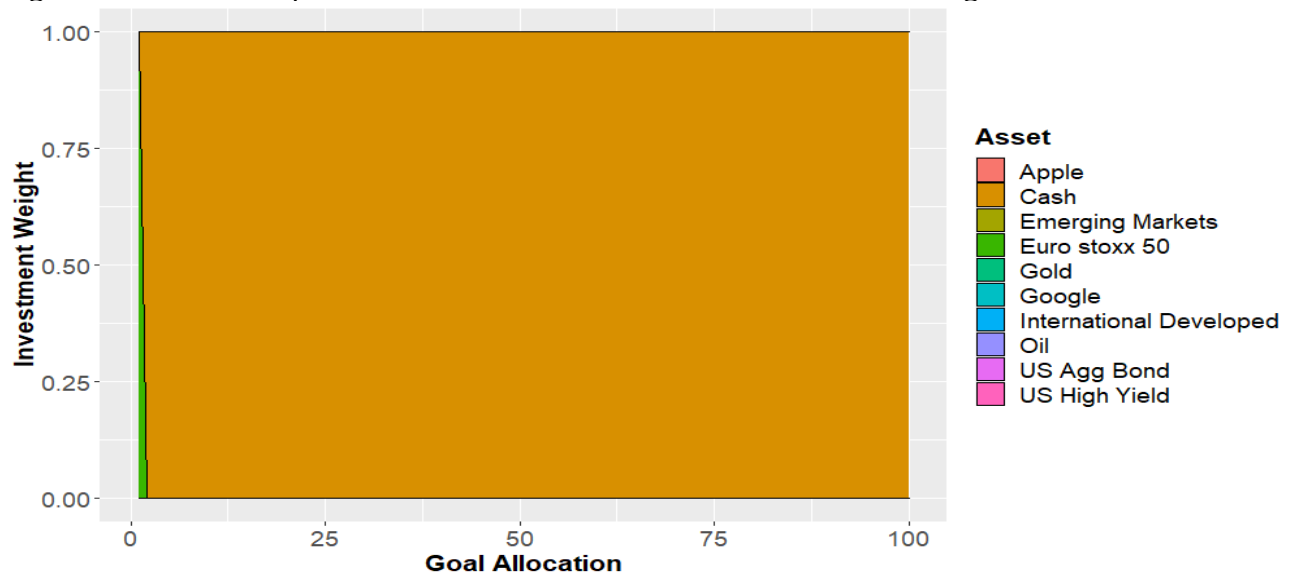
Source: R Studio

Figure 6- Goal C's subportfolio allocation as a function of Goal C's across-goal allocation



Source: R Studio

Figure 7- Goal D's subportfolio allocation as a function of Goal D's across-goal allocation

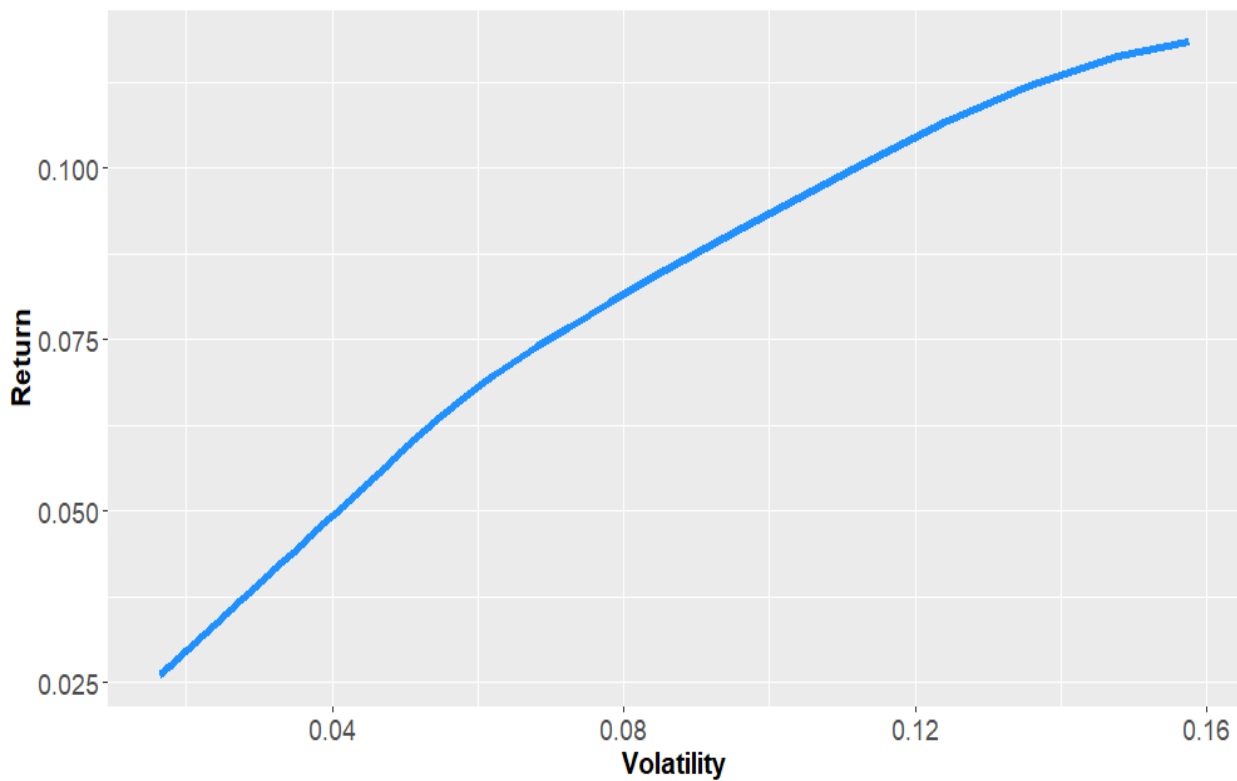


Source: R Studio

Mean- Variance Efficient Frontier

The optimal mean variance portfolios that offer the highest projected return for a given level of risk is presented in Figure 8. The Figure shows the portfolio of the goal-based investor who is mean-variance constraint. From the Figure, the Mean variance constraint goal-based investor has several positions on the frontier to choose from depending on his appetite for risk, that is the probability of failing to achieve a desired goal.

Figure 8- Mean- variance Frontier

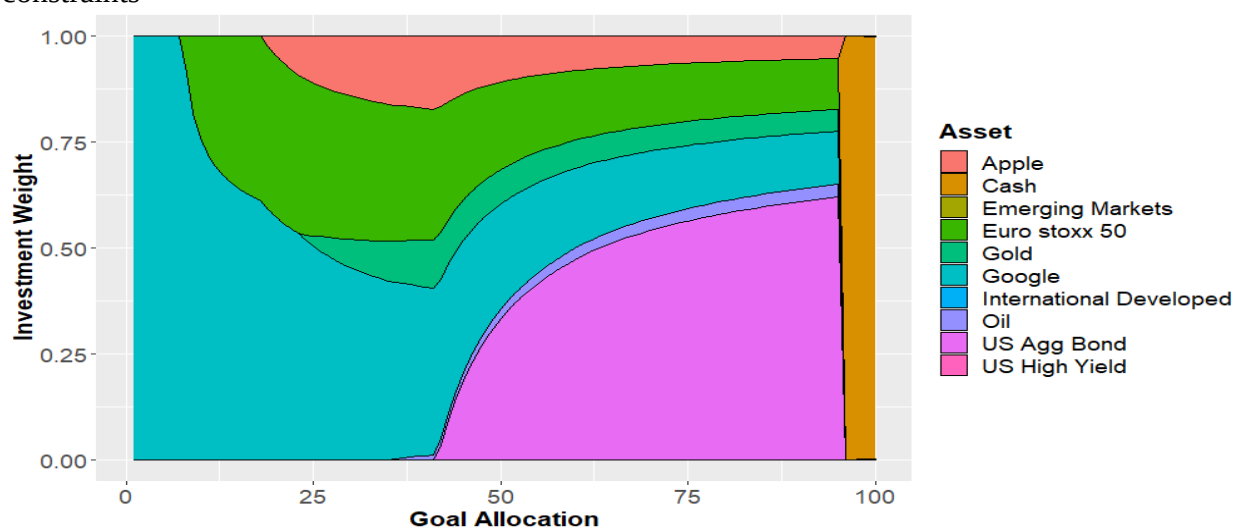


Source: R Studio

6.3 Investment Allocation as a function of Goal Allocation, with Mean-Variance constraints

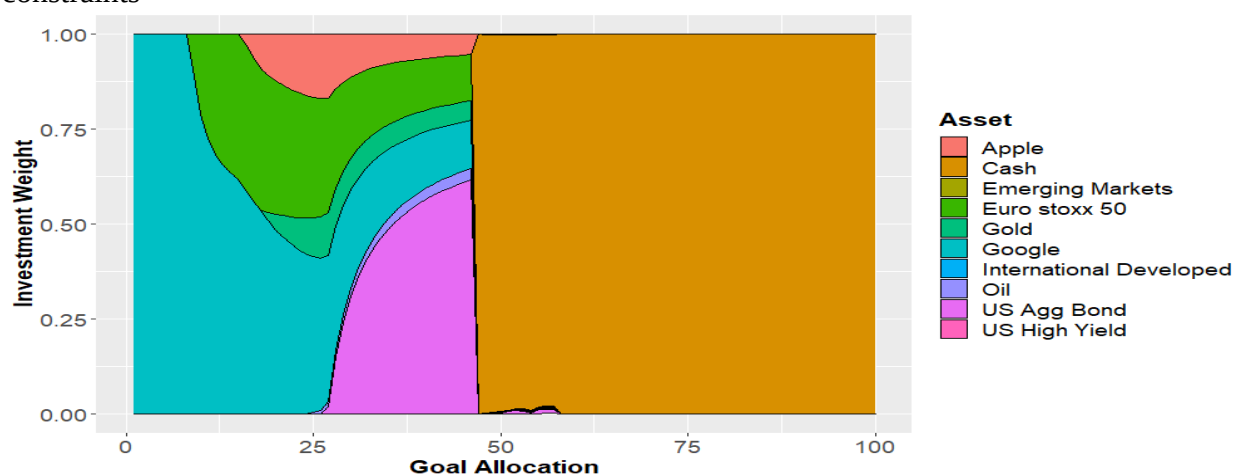
Figures 9,10,11, and 12 clearly depicts how the assets are allocated in an across goal specification, that is how one goal is substituted for another goal given mean-variance constraint. The asset allocation is almost similar as in the case of figure 4, 5, 6, and 7.

Figure 9- Investment Allocation as a function of Goal A Allocation, with Mean-Variance constraints



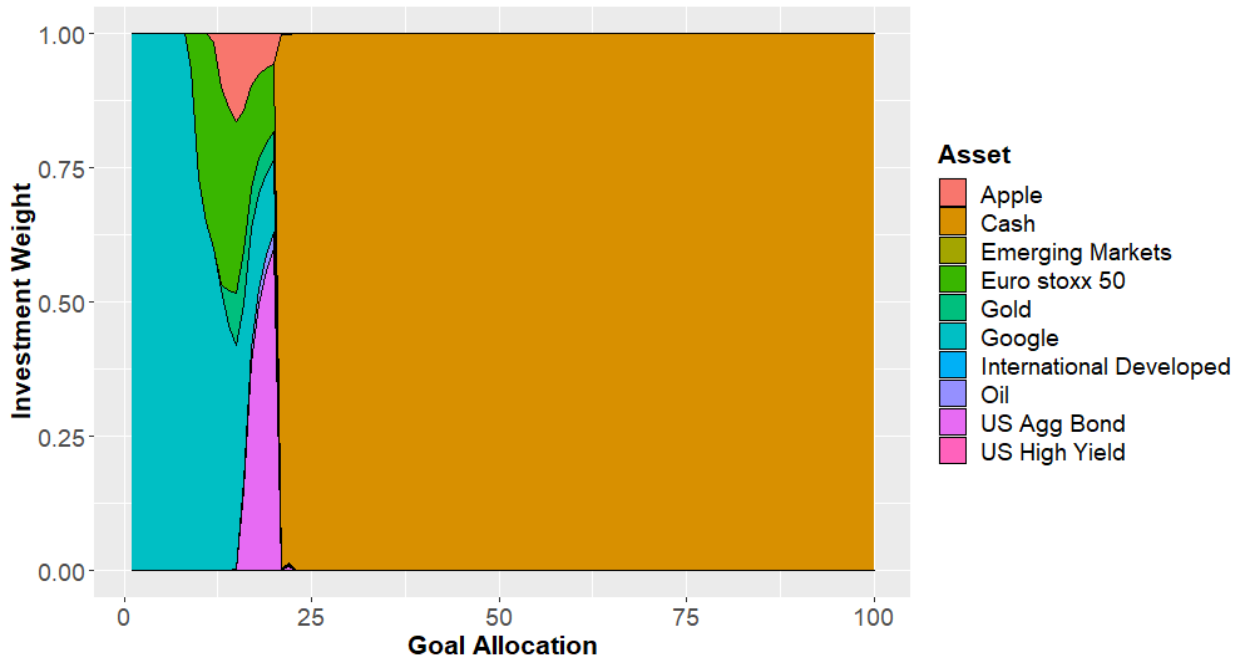
Source: R Studio

Figure 10- Investment Allocation as a function of Goal B Allocation, with Mean-Variance constraints



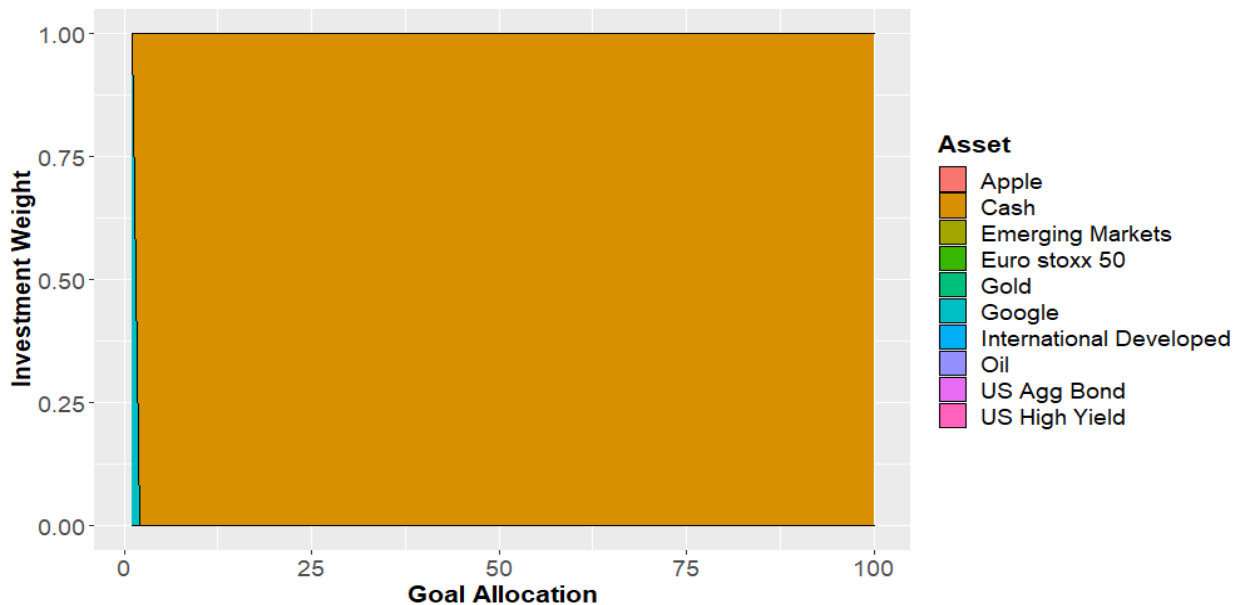
Source: R Studio

Figure 11- Investment Allocation as a function of Goal C Allocation, with Mean-Variance constraints



Source: R Studio

Figure 12- Investment Allocation as a function of Goal D Allocation, with Mean-Variance constraints



Source: R Studio

6.4 Probability of achievement for goals-based approach and mean-variance approach

The final aspect of the analysis is comparing the probability of goal achievement between the two approaches of portfolio modelling, that is Goal-based portfolio modelling and Mean-Variance portfolio modelling. The idea is to document which of the two approaches yields the highest probability of goal achievement. From Table 6.7, both approaches have the same goal weight. The findings are presented in the figures below

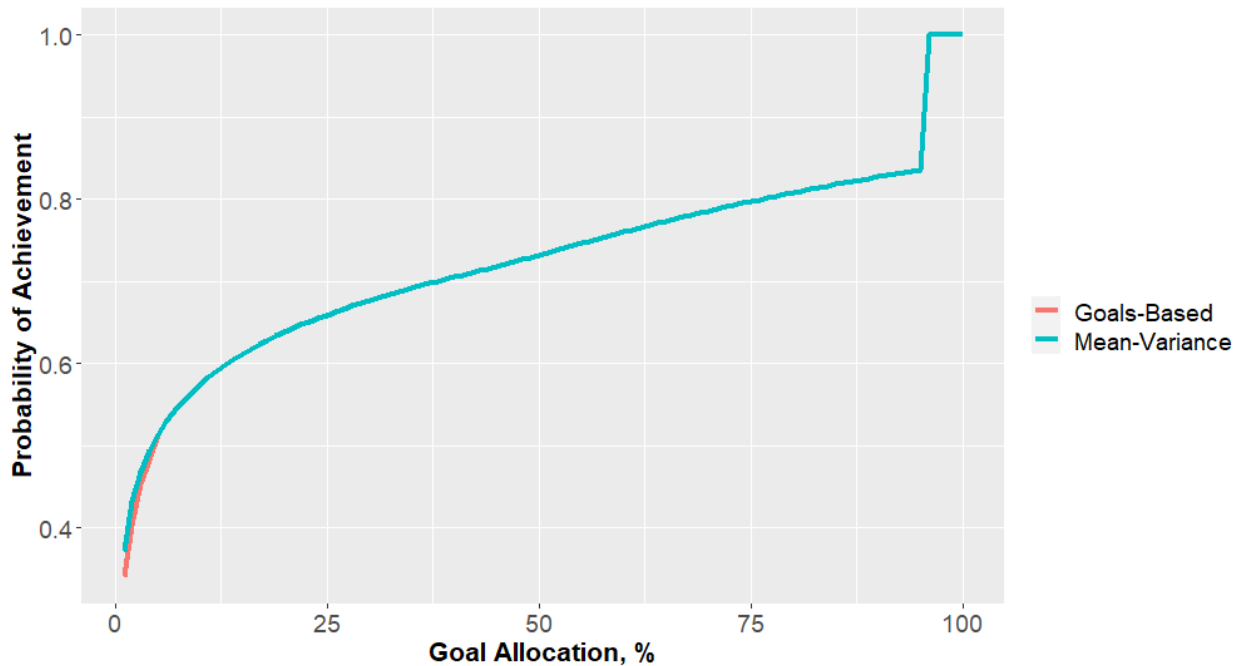
GOAL A

The optimal goal weight for Goal A as seen in Table 6.7 is 29. From Figure 13, the probability of achieving the goal based on the two approaches is 68%, This means that per the investment strategy set out for Goal A taking into consideration the initial capital and the capital market expectations, there is a 68% certainty of Goal A being achieved together with achieving other goals. This probability represents an across goal allocation which means the investor is concerned with the certainty of achieving all his stated goals.

An interesting observation can be seen from Figure 13. One can clearly see that for higher across-goal allocations, goal-based portfolios and mean-variance portfolios are the same. However, as wealth distribution to the goal drop, goal-based portfolios start to provide more probability of achievement than those offered by the mean-variance constraint portfolios. The point of departure between these two approaches is when a goal is valued too little or until there is not enough money to go around. In such instances, the MV solution and the GBI solution will part ways.

Mathematically, if the required return of the goal is higher than the last portfolio on the efficient frontier, then GB solutions will yield higher probabilities of goal achievement than MV-efficient solutions. One will therefore not be wrong in saying that mean-variance portfolios are stochastically dominated by goal-based portfolio.

Figure 13- probability of achievement of Goal A between the GB solutions and MV solutions



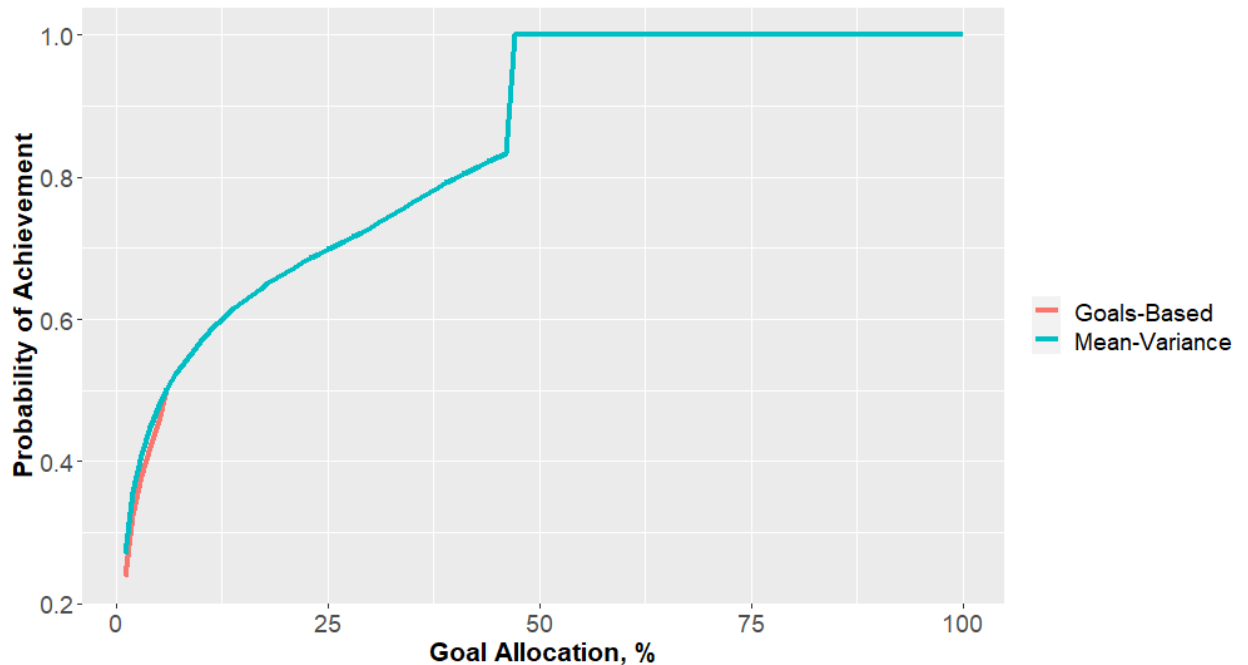
Source: R Studio

GOAL B

The optimal goal weight for Goal B as seen in Table 6.7 is 47. From Figure 14, the probability of achieving the goal based on the two approaches is 84%. This means that per the investment allocation design modelled for Goal B taking into consideration the initial wealth and the capital market expectations, there is a 84% certainty of Goal B being achieved together with achieving other goals. This probability represents an across goal allocation which means the investor is concerned with the certainty of achieving all his stated goals.

Also, from Figure 14, as it was in the case of Figure 13, it can be concluded that for higher across-goal allocations, goal-based portfolios and mean-variance portfolios are the same, that is provides the same probability of achieving a defined goal. However, as wealth distribution to the goal drop, goal-based portfolios start to provide more likelihood of achievement than those offered by the mean-variance constraint portfolios.

Figure 14- probability of achievement of Goal B between the GB solutions and MV solutions



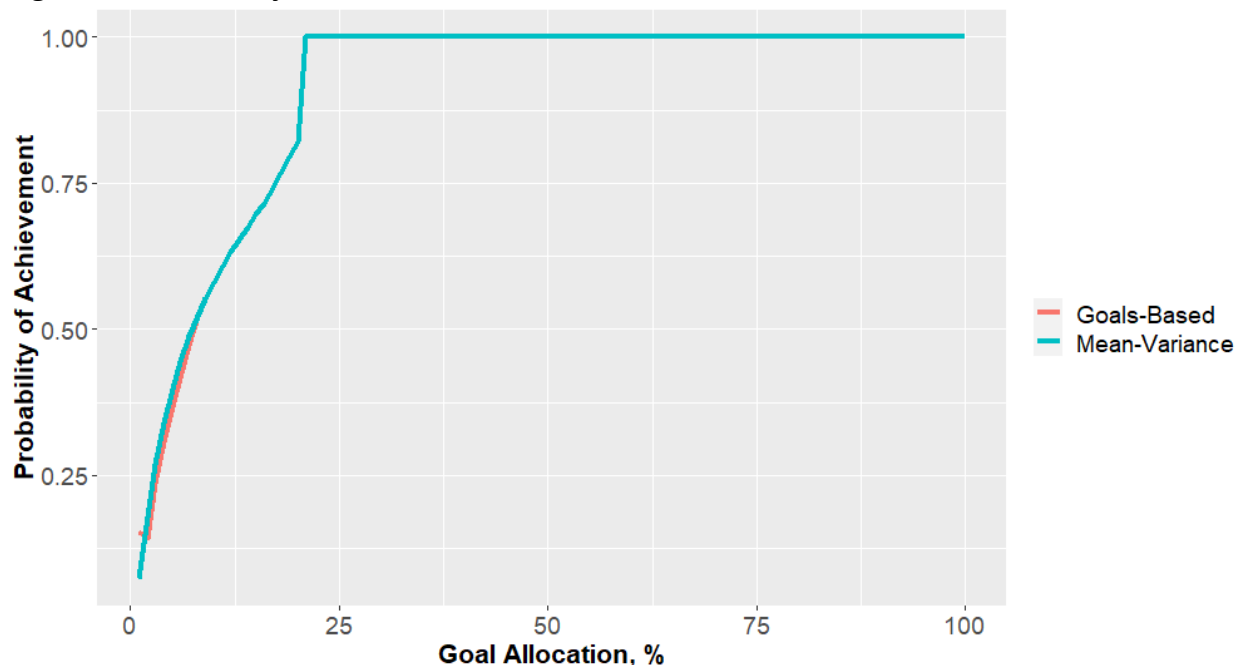
Source: R Studio

Goal C

The optimal goal weight for Goal C based on the two approaches as seen in Table 6.7 is 22. From Figure 15, the probability of achieving the goal based on the two approaches is 68%, This means that per the investment allocation design modelled for Goal C taking into consideration the initial wealth and the capital market expectations, there is a 68% chance of Goal C being achieved together with achieving other goals. This probability represents an across goal allocation which means the investor is concerned with the certainty of achieving all his stated goals.

Also, from Figure 15, as it was in the case of Figure 13 and Figure 14, it can be concluded that for higher across-goal allocations, goal-based portfolios and mean-variance portfolios are the same, that is provides the same probability of achieving a stated goal. However, as wealth distribution to the goal decrease, goal-based portfolios start to provide more likelihood of achievement than those offered by the mean-variance constraint portfolios.

Figure 15- Probability of achievement of Goal C between the GB solutions and MV solutions



Source: R Studio

Goal D

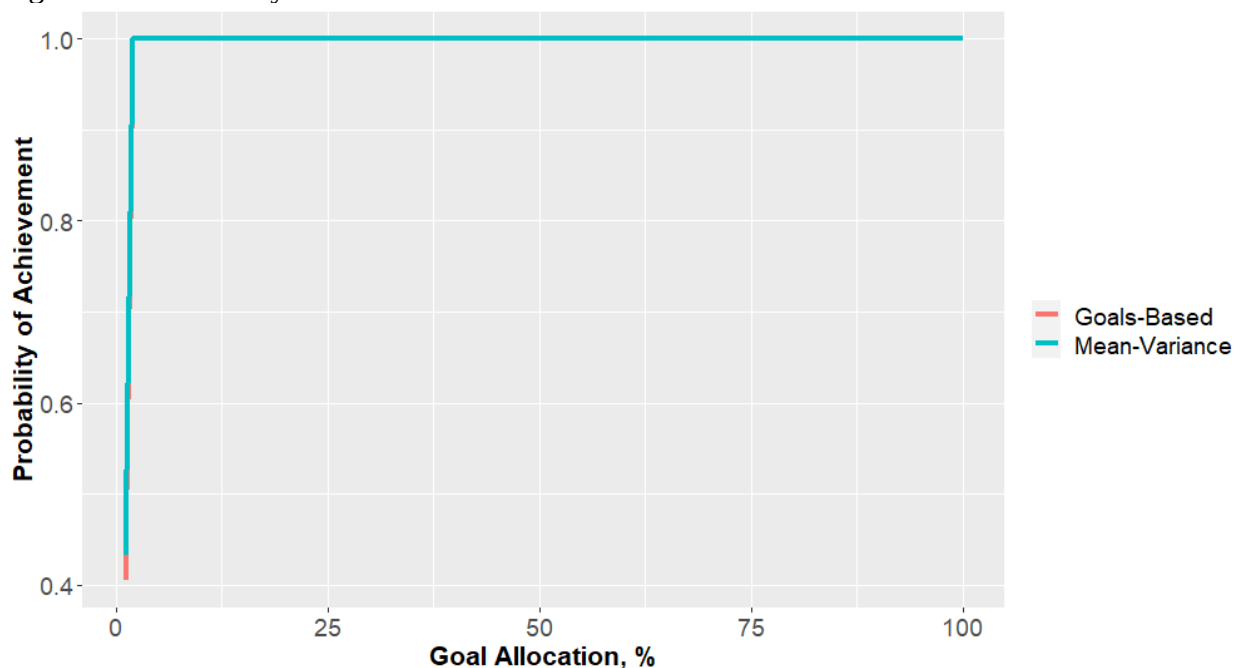
The optimal goal weight for Goal D based on the two approaches as seen in Table 6.7 is 2. From Figure 16, the probability of achieving the goal based on the two approaches is 100%. This means that per the investment allocation design modelled for Goal D taking into consideration the initial wealth and the capital market expectations, there is a 100% probability of achieving Goal D while achieving the other goals. The 100% probability means that in 5 years that the amount of \$70,000 that the goal-based investor is expecting to have to be able to go on the family vacation is guaranteed. This probability represents an across goal allocation which means the investor is concerned with the certainty of achieving all his stated goals.

This is the only goal that there is 100% degree of certainty in achieving.

Also, from Figure 16, as it was in the case of Figure 13, 14 and Figure 15, it can be concluded that for higher across-goal allocations, goal-based portfolios and mean-variance portfolios are the same, that is provides the same probability of achieving a stated goal. However, as wealth

distribution to the goal decrease, goal-based portfolios start to provide more likelihood of achievement than those offered by the mean-variance constraint portfolios

Figure 16- Probability of achievement of Goal D between the GB solutions and MV solutions



Source: R Studio

Actual future wealth

The study moved on to calculate the actual wealth the investor will have at the end of the investment horizon based on the probabilities of achieving the goals. The breakdown per goal is presented in Table 13. From Table 13, it is only Goal D that the expected future wealth is the same as the actual future wealth, all the other goals had their actual future wealth being less than the expected future wealth.

Table 13 – Expected future wealth vs Actual future wealth

Goal	Probability of Achievement	Expected Future wealth	Actual Future wealth
A	68%	\$4,500,000	\$3,060,000
B	84%	\$2,000,000	\$1,680,000
C	68%	\$800,000	\$544,000
D	100%	\$70,000	\$70,000

Source: Authors construct

Adding the personal bucket to the initial wealth (Liquidating residence)

From Table 13, it became apparent that the investor is below the threshold of probability of reaching his goals. From the onset, the investor indicated that he was willing to draw wealth from his personal bucket to add to the market bucket to help increase the probability of achieving the goals if the probability was below 90% for Goal A. The residence will then be liquidated at the initial date and the incomes will be re-invested in the market assets. This will result in a new starting weight for each asset in the portfolio. The new starting weight together with the wealth allocation are presented in Table 14.

Table 14- Asset starting weight after liquidating residence.

Asset	Starting weight	Wealth Allocation
Apple	5.41%	\$ 297,280.78
Eur 50	14.80%	\$ 814,022.44
GOOG	14.61%	\$ 803,630.25
AGG	16.77%	\$ 922,302.76
Gold	10.58%	\$ 581,908.80
Oil	5.35%	\$ 294,303.96
EM	10.87%	\$ 597,672.41
IFGL	5.96%	\$ 327,849.17
HYG	6.04%	\$ 332,078.95
CASH	9.62%	\$ 528,950.51

Source: Authors construct.

The new asset weight will lead to a new goal weight. The new goal weight after adding the residence of \$2,000,000 is presented in Table 15.

From Table 15, both approaches had the same goal weight as it was in the case of Table 11. From Table 15, Goal A now has the highest goal weight of 62 as compared to the initial goal weight of 29 in Table 11. Also, Goal B and Goal C's goal weight has dropped from 47 to 30 and

22 to 6 respectively. Goal D, however, maintained its weight of 2 despite the additional infusion of funds.

Table 15- optimal goal weight after liquidating residence

Goal	Optimal Goal weight (GB)	Optimal MV constraint weight
Goal A	62	62
Goal B	30	30
Goal C	6	6
Goal D	2	2

Source: Author's construct

Actual future wealth after adding personal bucket

Table 16 contains information on the new probabilities of goal achievement which was further used in calculating the Actual future wealth. It must be mentioned that these probabilities were computed in the face of the same capital market expectation, return forecast and correlation matrix used in computing the earlier probability. From Table 16, the probability of achieving Goal A now stands at 95% as compared to the initial 68% of achieving the Goal. The investor is willing to accept the 5% shortfall of not achieving Goal A than the initial 32%. Goal B maintained its original probability of 84% of achieving the financial goal of \$2,000,000 in 20 years. From Table 16, the probability of achieving Goal C has increased from 68% to 85% as it was in the case of Goal A. Goal D, like in the first instance is the only goal which will be achieved with certainty.

Table 16 – Expected future wealth vs Actual future wealth after adding personal bucket

Goal	Probability of Achievement	Expected Future wealth	Actual Future wealth
A	95%	\$4,500,000	\$4,275,000
B	84%	\$2,000,000	\$1,680,000
C	85%	\$800,000	\$680,000
D	100%	\$70,000	\$70,000

Source: Authors construct

6.5 Discussion

Individual and institutional investors mostly face mutually exclusive goals competing for scarce resources over differing time horizons. The daunting task is how the investor can rationally allocate resources across goal, that is substituting one goal for another goal (Parker, 2021, p.20). As explained earlier risk in the goal-based setting is the likelihood of not achieving a set of stated goals which is how investors intuitively perceive risk (Brunel, 2011).

The trust of the goal-based investing approach is to be able to solve investors lifetime needs by providing optimum lifetime asset allocation and as such it is only prudent that any new information is thoroughly captured to yield a revised ideal lifetime asset allocation (Madlina, 2016).

The study considered an investor with 4 different goals at different time horizons. With the help of a wealth planner (analysis) a mix of investment were proposed at the initial stage to evaluate the probability of achieving the stated goals. The 4 sub portfolios were constructed based on the Behavioral portfolio theory.

The investor allocated resources across the goal-space as a function of the comparative value of one goal in relation to another. Moreover, because each goal had its own vector of variables, the investor had to balance the optimal distribution within each goal.

From the analysis, it was evident that the dataset exhibited low, weak, high, and strong correlations among each other. This was very much needed as the negatively correlated assets meant that as one asset class is being affected negatively, the returns from the other asset class will make up for the negative returns from the other asset. This will ensure that the goal-based investor's portfolio is well diversified to stand the test of time which will eventually lead to an optimal performance in helping the investor achieve his stated goals.

The four different goals resulted in four (4) different sub portfolios. Each sub portfolio had its unique asset allocation predominantly based on the time horizon. Goal A which had the longest time horizon of 30 years had about 96% investment allocation in equities. The idea was to benefit from the rising prices of equities and to be able to withstand the shock of the capital market as it is expected that by 30 years' time, sub portfolio A would have been able to deal with the

volatility in the markets and still return the future expected wealth. This asset allocation is consistent with previous studies who also allocated most of their assets in equities for investment in long time horizon goal-based portfolio (Degust et al., 2021; Suryanarayanan et al., 2014). Realizing the initial wealth and investment allocation could not guarantee a higher probability of achieving Goal A, the residence was liquidated to infuse the proceeds to the initial capital. This did not change the investment mix, with only Gold gaining extra weight than the initial weight.

Sub portfolio B also had a similar asset weight allocation as Sub portfolio A. This was mainly due to its long-term investment horizon of 20 years. Sub portfolio C and D were rather conservative as they had shorter investment horizon resulting in selecting less risky assets which offered lower returns as compared to the equities.

From the analysis carried out, it was noted that the Current (initial) market wealth was too low to secure Goal A which was the most important goal of which the investor was not willing to accept more than 10% probability of failure. This finding is undebatable, as in the absence of arbitrage opportunities, it becomes increasingly impossible to achieve a goal with a higher degree of certainty regardless of the investment strategy implemented (Degust et al., 2021, p.127). As a practice with most goal-based approach, the investor had two options either to save annually to cushion his initial wealth or liquidate some of the assets in his personal bucket.

Aside the two options stated earlier, the investor has other well-known options also at his disposal namely, reprioritizing the investment objectives; reducing goal amounts; taking more shortfall risk; modifying the time horizon of goal accomplishment; or altering their asset allotment strategy (Taylor, 2019). Among the several options, the investor resorted to the option of liquidating some assets in his personal bucket by liquidating his residence to re-invest the proceeds in the market asset. This was a good strategy as it resulted in a 95% probability of achieving Goal A. When faced with a similar situation of not achieving a desired probability of success. Degust et al., (2021, pp. 134-135) rather resorted to savings, that is real annual savings. This increased the probability of achieving the most important goal from 78.6% to 95%.

For Goal B, the threshold of achievement was 83%. This was achieved with the initial resource pool of \$3,500,000. It is interesting to note that after increasing the initial wealth with the proceeds from liquidating the residence, the probability of achieving the goal did not change. It maintained the 84% probability of attaining the goal which the investor is prepared to accept. Although, Goal B is also important to Goal A, the investor is willing to accept a higher probability of failure than Goal A because the investor has several colleges, he is willing to choose from which can meet the 83% funding threshold. This colleges to the investor will each be liked by his child.

For Goal C, which was the goal with the least value ratio, the goal-based investor is willing to accept a higher shortfall than all the other goals. The threshold of achievement was slated at 80%. The initial wealth allocation resulted in a 68% probability of achievement which was below the threshold. However, after infusing the residence proceeds the probability of achievement increased to 83%. This finding is consistent with other studies (Taylor, 2019) in which the goal with the least goal value had the highest tolerance of risk, that is the probability of failing to achieve a goal.

Goal D maintained documented to be the only Goal with 100% certainty of fulfilment. The investor realizing the short duration of the investment decided to invest in safe asset (CASH) which had a lower volatility of returns.

Goal based investors are concerned with maximizing the chances of attaining the stated goals. For this reason, the study examined which of the two approaches, Goal-based investment approach and the Mean Variance approach could offer a higher probability of achievement in the lens of the framework developed by Das et al., (2010). From the analysis, as exhibited by Figures 13,14,15, and 16 it was realized that the goals-based technique stochastically dominates the MV approach. This finding is consistent with the findings of Parker (2021). Regardless of the MV approach being stochastically dominated by the Goal-based technique, investors deciding to remain Mean Variance efficient as exhibited by the Mean Variance efficient frontier was not the main trust of the larger discussion. The study was more concerned with allocation across goals (substituting one goal for the other) and less concerned about distribution across

investment simply because several investors are Mean Variance constrained. These two alternatives were both demonstrated during the analysis. A portfolio manager thus has the luxury to choose which alternative among the two options to go for, that is either going for probability maximization subject to mean variance constraint or adhering to precise probability maximization (Parker, 2021).

The findings of this study are consistent with previous studies. It affirms Boe et al., (2018) study which documented the long-term goal-based portfolio as having a lower probability of failure of 7% as compared to the traditional portfolio of having a 16% probability of failure. Also, it affirms Aleksandra (2020) study which revealed that investments based on a clear goal-based investing approach yields a higher probability of success in comparison with portfolios modelled on the traditional portfolio theory.

7 Conclusion

The aim of the thesis was to use an innovative approach to model a portfolio premised on the goal-based investing optimization approach. The study further sought to determine the probability of achieving the goal of the portfolio optimization. Moreover, the study analyzed how several optimized sub portfolios assigned to each goal in the goal-space performed. Lastly, a comparison was made between the probabilities of goal achievement in the goal-based space and the Markowitz Mean-variance space.

Analysis was carried out using data spanning from April 12th, 2010, to March 12th, 2022. The study modelled a robust goal-based investment approach to allocate resource across several mental account to account for the risk faced by the investor and how those risk is related to the investor's goals. The study presented a case study which can be considered as applications of the framework to various problems (investment goals) faced by investors. The portfolio was constructed based on the Behavioral Portfolio Theory.

From the analysis carried out, it was revealed that each sub portfolio assigned to a specific goal had its unique risk return characteristics leading to a unique set of investment mix and asset weight. Based on this result, it was concluded that sub portfolios can better serve the goals of an investor than the traditional portfolio theory which collapse multiple mental accounts into one portfolio.

It was further revealed after the initial analysis that the most important goal, retirement income's probability of achievement was below the threshold set out at the beginning of the investment process. Also, the third goal, that is the goal to purchase the house was below the threshold of accepted probability of achievement. However, the other two goal's probability of achievement met the accepted threshold. As with other practice, the study increased the initial wealth which resulted in all 4 goals achieving a higher desired probability of achievement.

From the analysis, it was further revealed that for higher across-goal allocations, goal-based portfolios and mean-variance portfolios offered the same probability of goal maximization. However, as wealth distribution to the goal drop, goal-based portfolios started to provide a

higher probability of achievement than those offered by the mean-variance constraint portfolios. It was therefore concluded that the goals-based framework technique stochastically dominates the MV approach at higher level of wealth distribution.

Goal-based investing is an upgrade of the traditional investment framework in how risk is measured and profiled while at the same time managing investors behavioral biases. The GBI framework must be embraced by all investment practitioners as a new way of framing portfolio as it is able to combine the older traditional finance framework with the behavioral biases framework to offer a maximum utility of goal achievement.

Recommendation based on findings

1. Investment advisors should be able to maintain mean-variance efficiency within goals while at the same time being able to efficiently distribute across goals.
2. Wealth planners must help investors develop their goal details and rank their goals in order of priority and assign the appropriate goal value.
3. Investment advisors must embrace the goal-based approach as it yields superior probability of goal attainment as compared to the mean-variance optimization approach

Recommendation for further studies.

Based on the findings from this study, the study recommends that:

1. further study should be carried on this topic using a different data set, that is a different investment mix, capital market expectation and goal funding details.
2. In place of increasing the initial wealth, further studies should examine the impact of savings on the probability of goal attainment.
3. Further Studies should be conducted considering the implementation constraints such as transaction cost.
4. Further studies should be carried out by allowing for short-sale and leverage constraint

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